European Single Market

How the programme has fostered competition

SUMMARY

The reduction in trade barriers in Europe through the Single Market Programme (SMP) was intended to increase competition in European markets, and hence welfare and efficiency. The paper examines how the SMP has affected trade patterns and what can be learnt from this regarding the impact on competitive behaviour. Combining an econometric and computable general equilibrium methodology, it is argued that (1) the SMP has indeed had a strong impact on competitive behaviour; (2) the extent of the impact depends both on changes in the intensity of competition as well as on changes in the nature of competitive interaction; (3) while all economies experience potentially large welfare gains from the SMP, it is the smaller economies that experience the larger gains, but this in turn suggests that they experience a greater degree of restructuring.

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Economic Policy October 1998 Printed in Great Britain
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The competition effects of the Single Market in Europe

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1. INTRODUCTION

The Single Market Programme in Europe was designed to generate structural changes in the European Union. The aim was that the elimination or reduction in trade barriers would reduce existing structural imperfections in European markets, and hence increase welfare and efficiency. Reducing trade barriers should in the first instance increase the intensity of competition and hence reduce price–cost margins. The increase in competition would in turn be likely to lead to industrial restructuring. Both the direct impact and any subsequent restructuring could in turn lead to greater exploitation of economies of scale. The trade barrier removal, the increase in competition and the potential expansion of scale would reduce distortions, leading to reductions in prices and thus an increase in welfare. It was also hoped that the Single Market Programme (SMP) would help to improve EU competitiveness in world markets.[1] In short, the aim of the SMP was to improve supply-side

This paper builds on work carried out for the European Commission's *Single Market Review* (1996). The support of the European Commission and its detailed comments and assistance with data provided for the original work are gratefully acknowledged. In particular, we would like to thank Peter Holmes, Daniel Gros, Paul Brenton and several members of the staff of the Commission's services for their comments. This paper also builds on earlier modelling work with Tony Venables, whose input and insights we gratefully acknowledge. Finally, the paper has been significantly improved following the helpful comments of the Economic Policy Panel, including Harry Flam, Peter Birch Sørensen and Klaus F. Zimmermann, although of course any errors and omissions remain our responsibility.

[1] Note that increased competitiveness arising from lower prices can also lead to terms of trade losses.

conditions across Europe by reducing the barriers to market access between countries.

There is increasing evidence that the SMP has indeed had a large impact. Aside from anecdotal evidence, the main *ex post* evaluation of the SMP derives from work commissioned by the European Commission (1996b). Section 2 below highlights some of the results, but the overwhelming conclusions from this review are that the SMP has had a large overall effect on competition, efficiency and hence welfare in Europe. What is striking, however, is that despite the volume of evidence presented, there has been no comprehensive study of the actual extent to which trade barriers have declined, and hence no attempt to separate out the consequences of the direct impact of the SMP from the subsequent changes in supply conditions induced by the trade liberalization. In other words, there has been little attempt to break down the overall impact of the SMP into the impact on patterns of production and trade (and welfare) arising directly from the lowering of trade barriers (the direct effect), and the resulting impact on price–cost margins and industrial restructuring (the competition effect).

It is this issue – the extent to which the SMP has impacted upon competitive behaviour within the EU – that the present paper addresses. In the paper, we examine how the reductions in trade barriers have affected trade patterns, both within the EU and between the EU and its trading partners; and then we look at what we learn from this about the impact of the SMP on competitive behaviour in European markets.

Methodologically, the paper combines econometric and computable general equilibrium (CGE) methodologies and provides a major step forward in analysing the impact of trade barrier reductions in two regards. First, the econometric model separates out the direct effects of the reductions in trade barriers from the impact of those reductions on competitive behaviour. Where previous work in this field (e.g., Winters, 1984) has focused only on the direct effects, our methodology allows us to distinguish between the demand and the supply side-effects of the SMP. The second feature of this paper is the way in which the econometric and the CGE analyses are linked: the econometric estimates of the actual impact of the SMP are used as inputs into the CGE analysis. Conventionally, CGE models use model parameters derived from (unrelated) econometric analyses: here policy parameters for our CGE model are derived from an econometric model with the same underlying economic structure, the same commodity disaggregation and common data.

The econometric approach seeks to make a clear distinction between the *direct* effects and the *competition* effects of the SMP by adopting a two-equation approach in a model of imperfect competition between firms operating in markets for differentiated products. In the first stage, the analysis identifies changes in the allocation of demand between domestic products, imports from other EU countries, and imports from non-EU countries, while controlling for income or price changes. In the second stage, the effects of the Single Market on price–cost margins are

identified, but controlling for labour and materials costs, for competitors' prices and market shares. The computable general equilibrium approach is based on the same economic ideas – imperfect competition with product differentiation – but with a radically different modelling strategy. This model also allows us to separate out direct effects and competition effects, and is more elaborate than that used in the econometric estimates; but it makes much less use of empirical data, which are used only to calibrate the model numerically, not to test it. To put it concisely, the econometric model is constructed in a way that takes account of theory, but is essentially driven by the empirical data, while the CGE model takes account of empirical data, but is essentially driven by the theory.

Our research is therefore based on two approaches with a common analytical underpinning and common data. As they are based on different methodologies, we should not expect them to give the same answers, but the extent to which the answers diverge gives us some sense of the extent of our true understanding of the impact of the SMP. Section 2 provides an empirical backdrop to the issues under consideration. Section 3 provides a brief theoretical and methodological discussion, and considers the ways in which the SMP could affect trade patterns and competitive behaviour. Section 4 discusses the econometric methodology and results. Section 5 outlines the computable general equilibrium model, which is then used to examine the competition effects of the SMP across 64 manufacturing sectors. Section 6 concludes.

2. OVERVIEW OF EMPIRICAL EVIDENCE

In the first detailed *ex post* analysis of the effects of the SMP, the 38 studies underpinning the European Commission's (1996b) *Single Market Review* provide strong evidence of the impact of the programme on European economies at both the micro and macroeconomic level.[2] The impact on price–cost margins, industrial restructuring and hence on competitive interaction emerges both from detailed sectoral studies as well as from broader-based studies.

For example, the available evidence on mergers and acquisitions suggests a considerable increase arising from the SMP. The intra-European share of world merger and acquisitions activity increased from 9.9% between 1985 and 1987, to 28.8% between 1991 and 1993 (European Commission, 1996a). Similarly, concentration in manufacturing industry (measured by the share in total sales of the largest four firms in each industry) has sharply increased. The unweighted average for total manufacturing increased from 20.5% to 22.8% between 1987 and 1993, and many industries experienced changes much larger than this (EAG (1996), based on Davies and Lyons (1996)). The most significant changes appear to have taken

[2] Good summaries of key results can be found in European Commission (1996a).

place in industries previously subject to public procurement distortions (e.g., aerospace), in food sectors sensitive to the SMP measures (e.g., pasta) and in certain electrical sectors (e.g., electrical machinery). There is also evidence of the impact of the SMP on the degree of competition in manufacturing industries as measured through changes in price–cost mark-ups, and through changes in price dispersion across EU countries (European Commission, 1996a). The estimates suggest a yearly reduction of 0.2 percentage points in mark-ups from 1987 onwards as well as considerable evidence of reductions in price dispersion arising from the SMP (DRI, 1996).

The SMP was intended to generate supply-side changes through internal trade liberalization. This led to concern in non-EU countries about the external impact of the SMP. The conclusions from the *Single Market Review*, however are that these concerns were unfounded. Indeed, as will be discussed below, it appears not only that the share of intra-EU imports increased as a result of the SMP, but also that the share of imports from the rest of the world increased substantially.[3]

In this paper, we examine how the change in trade costs has affected trade patterns and hence impacted upon competition in Europe. The standard way of assessing the impacts of integration on trade flow is to focus on the changes in shares of apparent consumption. These can be divided among three sources of supply: the share of domestic production in domestic consumption (Home), the share of imports from the EU (M_{EU}), and the share of imports from the rest of the world (M_{ROW}). Table 1 categorizes the effects of integration on trade patterns in terms of impacts on these shares. A blank entry implies no change in the share, a positive sign an increase, and a negative sign a decrease.

If imports from a partner country replace local production, this is (internal) trade creation and we have the changes in shares shown in the first row, while the second row corresponds to the case where imports switch from third-country to partner-country sources: (internal) trade diversion. The last three lines refer to the effects on external trade barriers. Integration could reduce such trade barriers – for example,

Table 1. Direct effects of economic integration

Case	Description	Home	M_{EU}	M_{ROW}
1	Trade creation	–	+	
2	Trade diversion		+	–
3	External trade creation	–		+
4	External trade diversion		–	+
5	Trade suppression/erosion	+		–

[3] It is interesting to note, however, that in order to assess the impact of the SMP many of the above studies have resorted to *ex ante* estimates of the extent to which trade barriers were reduced. The main source for this is almost invariably Buigues *et al.* (1990).

if a shift from a free trade area to a customs union led to the implementation of a low common external tariff in a country that formerly had high tariffs – and this could give rise to external trade creation and diversion, as domestic production and partner imports are replaced by foreign imports. On the other hand, high external barriers could have the reverse effect to external trade creation – trade suppression.

Figure 1 details the evolution of these shares over time for the EU12 for manufacturing in aggregate, while Figure A1 in Appendix A details the changes for each of the Nace two-digit industries. These figures shed light on whether any changes in the shares from the early 1990s onwards are part of a more long-term trend, or might be attributable to the SMP. It is important to note that in 1993 there was a change in the reporting system for trade flows which is likely to have impacted on the data. In the figures, therefore, we report both on the actual shares and on shares that have been adjusted to take account of these changes in the reporting of trade.[4]

From Figure 1 it can be seen that since the early 1980s there has been a steady trend towards greater import penetration with home shares declining and both EU

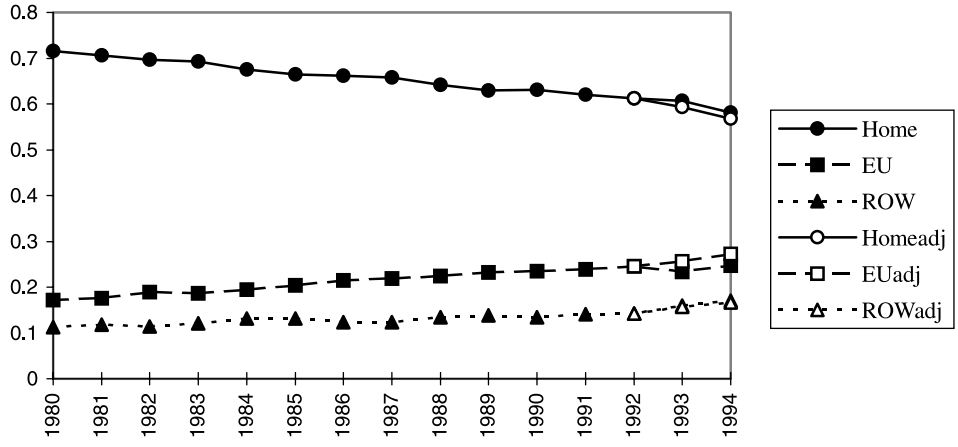


Figure 1. EU shares in apparent consumption – all manufacturing

[4] The adoption of the Intrastat system in 1993 means that intra-EU trade flows are reported by companies rather than customs officials. It is generally believed that consequently there is systematic underrecording of intra-EU trade. Two effects can be identified: underreporting of 1992 trade flows, which were subsequently reported in 1993 (a switch in the allocation of the reporting of trade flows across years); and a generalized underreporting of intra-EU trade flows following the adoption of the new system. Clearly, the former effect would have only a short-term impact and the use of data up to 1994 should overcome this. With regard to the second effect, work in this area done by the UK Central Statistical Office (CSO) (Williamson and Porter, 1994) suggests that the situation is complex. Adjustments made to initial estimates of imports in 1993 are worth about 1% of overall EU imports. In addition, the data prior to 1992 have been adjusted downwards by some 3.5% due to imports from the rest of the world through Rotterdam being improperly classified as EU imports. Unfortunately, the CSO was not able to allocate such trade on an industrial level. The total net effect of the UK CSO adjustments would appear to be to change the relative balance of EU and non-EU imports between 1992 and 1993 by some 8% of total EU imports. On this basis in Figure 1 we show what the shares would be if the figure for intra-EU trade were increased by 10% in 1993 and 1994.

and rest of the world (ROW) shares rising. The late 1980s saw a slight levelling off of the EU share, which then appears, on the basis of the adjusted shares, to have marginally increased in the 1990s. Looking at the shares by sector, the evidence is mixed. There are sectors where EU import shares do seem to have risen over the period in question (Nace 25, 26, 34, 35, 37); however, there are also industries where the change in shares appears to be simply part of a longer-term trend. Note that interpretation of the evolution of shares is sensitive to the adjustment we have made, where in some sectors (Nace 22, 24, 25, 26, 34, 43A) the adjustment irons out anomalies; whereas there are other sectors (Nace 23, 31, 32) where a counter-intuitive fall in the EU share remains even after the adjustment.

It is also clear that EU markets have become increasingly open to imports from the ROW, with changes in the 1990s being the continuation of a longer-run trend. The chemicals sector (25) is a clear example, in which the intra-EU import share has risen steadily too, and motor vehicles (35) and food, drink, tobacco (41) show the same changes, with man-made fibres (26) only a little less regular in the pattern of change over time. Metal manufactures (31), mechanical engineering (32), textiles (43A) and paper (47) show an increasing pace of ROW import penetration. Office machinery (33), other transport (36) and instrument engineering (37) all show a speeding up in the 1990s of the decline in home market shares, with the clear suggestion that the SMP or other changes in the 1990s have increased the openness of markets to imports from the ROW as well as the EU. There is also some, less strong, indication of an increase in the trend to openness in 34 (electrical engineering).

Clearly, changes in shares can also take place due to changes in exchange rates. We discuss this issue more fully in Allen *et al.* (1996). However, it does not appear likely that these results are driven by exchange rate changes. Where the rest of the world share has increased in all cases, this is not represented by a significant change in the late 1980s and early 1990s, but appears to be part of a longer-term trend. This would suggest that it is taking place as a result of the increasing openness of the world trading system, which is serving to reduce the domestic market shares at the expense of both rest of the world and EU producers. In addition, the increasing intra-European openness arising from the SMP might have further increased the attractiveness of the European market for rest of the world producers.

3. TRADE LIBERALIZATION AND COMPETITION EFFECTS

The effects of integration on patterns of trade and production can be seen in three phases (Baldwin and Venables, 1995). The first phase, the traditional theory of customs unions (Viner, Meade, Johnson), was based on the perfectly competitive comparative advantage trade model; the second phase stemmed from developments in the international trade literature of the late 1970s and early 1980s, and focused on the importance of imperfect competition and economies; the third phase

introduced the dynamics of the effects of integration on saving, investment and economic growth, and arose from the analytical developments of the late 1980s (Romer, Lucas and others).

As the first phase is based on perfect competition, integration does not as such yield changes in competitive behaviour. Sectoral restructuring occurs in response to the direct effect on prices of changes in trade barriers along traditional comparative advantage lines. The traditional approach recognized that there could be other effects (e.g., the work of Corden on economies of scale), but a satisfactory treatment of such effects became possible only after the development in the late 1970s of models of trade with imperfect competition (Krugman, Dixit and Norman, Lancaster).

In this literature, imperfect competition either can be the underlying cause for international trade (as in the reciprocal dumping model) or is needed in order to allow for product differentiation and economies of scale. By opening up competitor markets, trade liberalization is likely to increase the degree of competition. If one also allows for the entry and exit of firms, the consequent restructuring can allow remaining firms to exploit economies of scale further. In a world of imperfect competition where firms have market power, price is greater than marginal cost. The extent to which price is greater than marginal cost will then depend on the degree of concentration in the industry, on firms' perceived elasticity of demand for their products, on the extent of economies of scale in the industry, and finally on the nature of competitive interaction among firms. Economic integration can clearly impact on each of these and thus have substantial effects on the participating economies.

Important also is the distinction between segmented and integrated markets. When there is national market segmentation, firms set prices in each national market independently. Integration will generate increased intra-union trade as internal liberalization cuts price–cost margins within the union and leads to larger-scale production by fewer firms. Note that in terms of the taxonomy of Table 1, the creation of intra-industry trade need not generate increased net imports, as increased gross imports in any one sector for a particular country could be more than matched by increased gross exports to partner countries in the same sector. However, where the SMP also leads to the elimination of market segmentation, producers treat the union market as a single market in which they have to adopt a unified pricing strategy. This represents a much stronger pro-competitive impact. The reduction in price–cost margins across the union makes partner-country markets relatively less attractive than they are under market segmentation, so there will be less intra-industry intra-union trade: the effects on trade creation may be negative. Much of the discussion of the SMP has focused on the effects on competition of such barriers, notably the way that regulatory differences and public procurement practices can facilitate market segmentation (e.g., Smith and Venables, 1988).

With both segmented and integrated markets, the competitiveness of union-based producers may generate trade diversion as foreign-based producers lose market share in union markets. However, there is also the possibility of the market-access effect analysed by Smith and Venables (1991): the unification of regulatory systems within the union may make union markets more accessible to foreign producers, so giving rise to external trade creation.

Turning finally to dynamic or intertemporal effects, if integration leads to increased income in the union and therefore to increased saving and investment (Baldwin, 1990; Ben David, 1993), trade with both partners and non-partners should increase, but not at the expense of domestic production. There seems no reason to suppose that dynamic effects will give rise to any particular pattern of change in consumption shares, and indeed it is plausible that different patterns would be observed in different sectors if, for example, endogenous growth led to the expansion of capital-intensive and high-technology sectors and the decline of low-skilled and low-technology sectors.

Reductions in non-tariff barriers such as regulatory costs on cross-border transactions are at the core of the SMP, and the direct effect of such changes is the natural starting point of analysis of the programme. However, the central purpose of the SMP is to remove regulatory obstacles to competition, allow firms to expand their markets without monopolizing them, and enlarge the variety of choice available to consumers. The remainder of this paper focuses on and separates out these direct and competition effects.

4. ECONOMETRIC ANALYSIS

This section presents the results of a detailed econometric analysis of the impact of the Single Market Programme in a group of narrowly defined industrial sectors. The econometric model consists of a structural two-equation model of demand and supply in an imperfectly competitive industry. The model is estimated for fifteen sectors in each of four countries of the European Union (Germany, France, Italy and the United Kingdom). Full details of the results appear in Allen *et al.* (1996). The fifteen sectors were chosen on the basis of their *ex ante* identification by Buigues *et al.* (1990) as being particularly sensitive to the Single Market Programme. Together the sectors make up some 36% of total EU manufacturing output. For comparison purposes, the same analysis was performed on data for manufacturing industry as a whole.

The main innovation of our work in comparison to earlier work such as that of Winters (1984) is in the examination of the supply-response effects, which is done within the framework of an explicit structural model of imperfect competition with economies of scale (Allen, 1994). Price mark-ups over marginal costs are assumed to depend on market share. The larger a firm's market share, the more inelastic its own-price demand, and hence the greater the scope for high price–cost margins.

The model therefore allows both for economies of scale and for a supply function that is upward sloping in terms of market share. Analysis of the theoretical model shows that the firm supply function is flatter, the higher the degree of competitiveness of the industry and the higher the degree of substitutability of products.

4.1. The theoretical framework of the econometric analysis

The theoretical framework can best be described in partial equilibrium terms. This is portrayed in Figure 2, which is a standard demand–supply diagram for domestic production and consumption, except that the horizontal axis measures domestic *market share* rather than volume of sales. Note that the supply curve is upward sloping, *not* because of increasing costs, but because domestic firms' market power – and hence ability to increase price–cost margins – increases with their market share.

The overall impact of trade barrier reduction can then be decomposed into the following three components.

(1) *Direct demand effects.* A demand system is estimated to explain the behaviour of domestic, EU importer and outside-EU importer market shares in terms of relative prices and income effects. A set of dummy variables is included in these equations over the period of the Single Market Programme. These dummy variables therefore capture unexplained shifts in market shares, conditional on prices and real income, which we attribute to Single Market effects. The direct demand effects can therefore be seen as a shift in the demand curve, holding prices (and income) constant. In terms of Figure 2, this effect is shown by the move from *a* to *b*.

(2) *Competition effects on price–cost margins.* The impact of increased competition on price–cost margins is derived from the estimation of a domestic price equation. This is done using an explicit model of imperfect competition, where domestic firms' prices are explained in terms of labour and materials costs, competitor prices and the level of output relative to trend (in order to take account of the effect of the

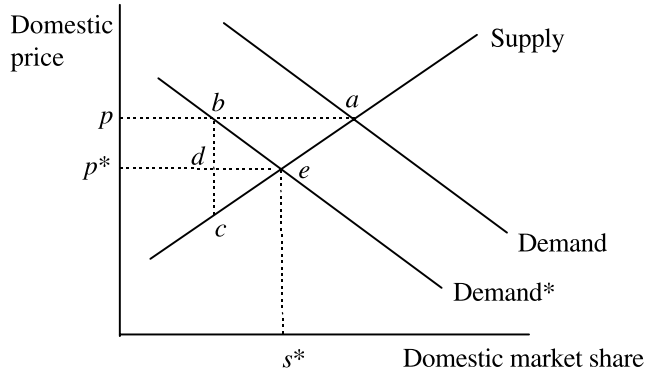


Figure 2. Demand and competition effects

normal business cycle). Once again the SMP effects are described by a set of dummy variables which represent the effect of the change in domestic market share on price–cost margins.

In Figure 2 (holding marginal costs constant) the change in market shares engenders a fall in price–cost margins in the direction of *c*. However, changing prices of course impacts on demand such that the final equilibrium is achieved at *e*. The actual reduction in price–cost margins as a result of the loss of market share is thus the distance between *b* and *d*. The fall in overall domestic market share is therefore attenuated by the distance from *d* to *e*.

(3) *Overall effects of the Single Market Programme*. The overall partial equilibrium effects of the Single Market Programme must take account of both the above demand and supply responses. The competitive price-cutting response of domestic firms will to some extent restore the loss of market share caused by the direct demand effects of the Single Market Programme. The scale of the overall Single Market effect has been derived through simulation of our estimated demand and supply equations. The overall effect is therefore measured by the shift back down the supply curve from *a* to *e*.

The data used were derived from Eurostat industrial databases together with macroeconomic indicators from the OECD Main Economic Indicators. The sample period covered 1976–94 using annual data. Fifteen three-digit NACE industries were studied in Germany, France, Italy and the UK, selected as described above. The detailed sectoral data available included the current value of domestic sales, intra-EU imports and outside-EU imports, together with a set of industry-level domestic producer price indices. Data on labour costs and intermediate input prices were not consistently available at the sectoral level; we have therefore used the corresponding macroeconomic time series, available from the OECD. The period of the SMP is taken as 1992–4, with a dummy variable added to each of the equations for each separate year over that period. This is clearly a minimal definition of the Single Market period chosen because of the particular interest in the impact of the programme over this time period. With a relatively short sample period, it also made sense to maximize the power of the structural break tests by choosing *a priori* a break-point. Sensitivity analysis revealed broadly similar, though quantitatively larger, results for longer periods (see Box 1). This, together with the overall consistency and scale of the effects found in our results, suggests that the choice of the treatment period gives a minimum measure of the impact of the SMP.

The SMP effects have been measured as the permanent shift in the constant term of the long-run or static equation relationship, i.e. after dynamics have been taken into account. The econometric equation dynamics are therefore fully taken into account in deriving the cumulative effects of the dummy variables.

The estimated demand equations were derived from a fully theoretically consistent almost-ideal demand function with a partial adjustment form. The shares of domestic sectoral expenditure on domestically produced goods, intra-EU and

Box 1. The dynamics of the Single Market Programme: the example of pharmaceutical products

The exact timing of the impact of the Single Market Programme is a difficult question to assess. We have taken the period of impact of the programme as the period 1992–4. This can be seen as a minimal definition of the time period of the Single Market, which was initially announced in 1986. The latter period was taken because there is some evidence that Commission directives only actually passed generally into national legislation as the 1 January 1993 deadline approached. Secondly, this is the period of the removal of border controls. To examine the extent to which Single Market effects occurred earlier than our treatment period, we have examined in more detail the experience of the pharmaceuticals sector.

Pharmaceutical products (Nace 257) is a sector dominated by highly regulated markets and public procurement. It is a highly concentrated industry with price controls in many countries (see Emerson *et al.*, 1988). It was placed high among the 'most sensitive sectors' of the Single Market Programme discussed by Buiges *et al.* (1990).

The Single Market process is slow and protracted. For pharmaceuticals, the 'Transparency of Medicines' directive, which required governments to take quicker decisions on prices and reimbursement of medicines and to provide an explanation of decisions taken, was proposed in December 1986 but not finally approved until 1989. It subsequently had to pass into national legislation. A subsequent 'Rational Use of Medicines' directive was approved in 1992 to create common rules on drugs advertising and consumer information.

To test the timing of the effects of the Single Market Programme on pharmaceuticals, we introduced dummy variables into our equations for the periods 1988–94 and 1990–4 onwards to compare the results with those of the chosen treatment period 1992–4. The results make clear that the principal effects occurred over the period 1992–4, although there is evidence of some effects over earlier periods.

Examining home demand share effects first, the average reduction of home market share over the period 1988–94 was −0.8 percentage points (*t*-statistic −1.33), rising to −1.0 percentage points (*t*-statistic −2.24) in the period 1990–4, and to −1.9 percentage points (*t*-statistic −2.33) over the 1992–4 period.

The effects on price–cost margins are even more striking. The average reduction in the price–cost margin over 1988–94 was actually positive, over the period 1990–4 it was negative by 0.9% (*t*-statistic −1.00), while the fall in price–cost margins over the 1992–4 period was −7.0% (*t*-statistic −6.51).

extra-EU imports were explained using relative price indices and total real sectoral expenditure. The domestic price variable was the domestic sectoral producer price index. The prices of intra- and extra-EU imports were measured by Paasche indices of individual exporter country sector-specific domestic producer price indices, weighted using 1990 country market shares.

The estimated price equations took account explicitly of both competitor prices and marginal costs. Marginal costs were derived from a Cobb–Douglas cost function including labour costs and other input prices. Competitor prices were weighted on 1990 country market shares. A partial adjustment formulation of these equations was employed.

4.2. The econometric evaluation of the impact of the Single Market Programme

The principal empirical results of the study are presented in Table 2 and Figure 3. The table decomposes the overall effects of the SMP on market shares into direct demand-side effects and the effects of increased price competition. Columns (1)–(3) give the direct demand effect, columns (4)–(6) detail the further changes arising from the competition effects, and columns (7)–(9) the overall (net) effect. The figures in this and subsequent tables and diagrams represent the unweighted averages of the estimated SMP responses in Germany, France, Italy and the UK.

The direct demand impact results suggest that the Single Market has been trade creating, for both EU and non-EU producers. There is no evidence of any substantial trade diversion of non-EU trade. EU and non-EU producers have in fact gained roughly in the same proportion. This strongly suggests that the market-orientated measures of the SMP have opened up EU domestic markets to both EU and non-EU exporters. After allowing for other influences on trade patterns, the average domestic market share has fallen by 5.4 percentage points in the fifteen sectors studied, and by 2.2 percentage points in manufacturing as a whole. Domestic market share is estimated to have fallen in all of the sensitive sectors.

As far as price convergence is concerned, there is evidence that the Single Market has increased price competition. The results for the different sectors are presented in column (10) of the table. On average, after allowing for other influences on pricing, price–cost margins have fallen by 3.9% since 1991 across the fifteen sectors studied. This fall is reflected in all but one of the sectors studied. In manufacturing as a whole, there is a similar-sized reduction.

One of the interesting results of the study is the difference in response between different industries. This gives us some handle on the nature of the supply curves in different industries. Figure 3 shows the relationship between the overall (simulated) changes in market shares and the fall in price–cost margins. There is little direct correlation between the size of the estimated direct demand effects and the estimated competition impact. Industries have clearly adjusted in different ways.

Table 2. Econometric estimates of the impact of the Single Market Programme (% point change in market share)

	Estimated direct demand impact			Estimated competition impact			Estimated overall impact			Change in price–cost margins (%)
	Home (1)	EU (2)	ROW (3)	Home (4)	EU (5)	ROW (6)	Home (7)	EU (8)	ROW (9)	(10)
247 Glassware	−1.30	−0.10	1.40	0.70	−0.10	−0.50	−0.70	−0.20	0.90	−4.0
248 Ceramics	−4.20	1.80	2.40	−0.20	0.30	−0.10	−4.40	2.00	2.40	−3.0
251 Basic industrial chemicals	−4.30	2.50	1.80	1.10	−0.70	−0.30	−3.30	1.80	1.50	−6.2
257 Pharmaceuticals	−1.90	0.40	1.50	−0.10	0.20	−0.10	−2.00	0.50	1.40	−7.0
315 Boilermaking etc.	−5.30	4.40	0.90	0.90	−0.90	0.00	−4.40	3.50	0.90	−4.8
322 Machine tools for metals	−2.00	−2.20	4.20	−0.60	0.20	0.40	−2.60	−2.00	4.60	−6.7
324 Machine tools for foodstuffs	−7.40	3.00	4.40	0.50	−0.40	−0.10	−6.90	2.60	4.30	−1.4
325 Plant for mines	−1.70	1.00	0.70	1.10	−0.60	−0.50	−0.60	0.40	0.20	−4.3
330 Office machines	−7.80	2.80	5.00	1.10	0.10	−1.20	−6.70	3.00	3.80	−15.7
344 Telecommunications equip.	−2.70	1.70	1.00	1.00	−1.50	0.50	−1.70	0.20	1.50	−4.6
345 Electronic equipment	−15.70	4.60	11.10	4.00	−2.20	−1.80	−11.70	2.20	9.50	−1.1
351 Motor vehicles	−4.90	3.70	1.20	0.30	−0.70	0.30	−4.60	3.00	1.50	+2.2
364 Aerospace equipment	−15.30	14.60	0.80	7.00	−2.50	−4.40	−8.30	12.00	−3.80	−6.6
427 Brewing and malting	−6.30	5.90	0.40	1.50	−1.40	−0.10	−4.80	4.50	0.30	−0.1
453 Clothing	−2.90	−2.50	5.40	0.70	−0.50	−0.20	−2.10	−3.10	5.20	−1.8
Average of 15 sensitive sectors	−5.40	2.95	2.45	1.22	−0.80	−0.42	−4.20	2.13	2.03	−3.9
Rest of manufacturing	−0.42	−0.86	1.28	−0.83	0.44	0.39	−1.25	−0.41	1.67	−3.4
Total manufacturing	−2.20	0.50	1.70	−0.10	0.00	0.10	−2.30	0.50	1.80	−3.6

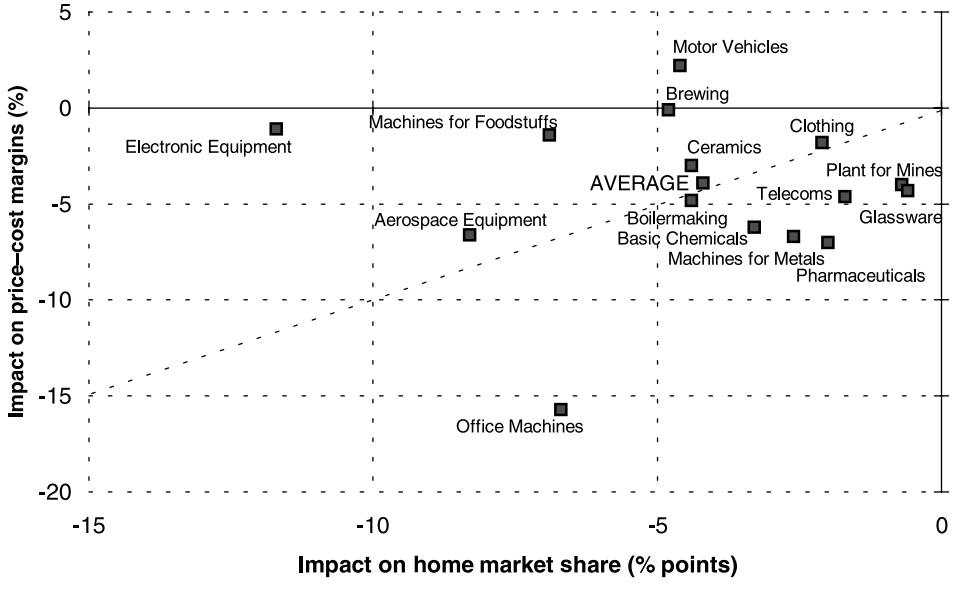


Figure 3. Effect of SMP on home market shares and price–cost margins

5. A COMPUTABLE GENERAL EQUILIBRIUM ASSESSMENT OF THE SMP

The CGE model operates at the same three-digit level as the econometric model, but with more comprehensive sectoral coverage than the econometric model, and it too treats producers as imperfectly competitive firms operating in markets with differentiated products. However, as a fully specified general equilibrium model of trade and production, the CGE model is more elaborate than the econometric model, allowing a richer set of linkages between markets – notably, linkages between sectors through intermediates and through the labour markets (see Box 2 for a description of CGE methodology).

Given the greater analytical complexity of the CGE model, we proceed as follows. First, we use the results from the econometric analysis in order to determine the policy parameters (change in trade costs) that would generate the change in shares that in Table 2 are attributed to the direct demand effect of the SMP. We then allow firms to adjust their price–cost margins in the face of increased competition, and here we allow for both segmented and integrated markets. We assess the impact on traded shares and examine the extent to which the CGE model arrives at results that are comparable to the econometric. In order to ensure consistency with the econometric model, the above involves running the CGE model in virtually a partial equilibrium manner (for example, through holding factor prices, intermediate input prices and firm scale constant). We then relax these assumptions and allow for the full interplay of general equilibrium interactions. This generates what might be thought of as the long-run impact of the SMP.

5.1. The model

The details of the model are outlined in Appendix C to this paper, and the full model is given in Gasiorek *et al.* (1992), who use the same basic model, at a more aggregated level. The underlying theoretical model, just like our econometric model, is based on imperfect competition and increasing returns to scale.

The model has twelve countries: each of the EU countries (with Belgium–Luxembourg treated as a single country) plus the rest of the world. Each country is endowed with three primary factors of production – capital, and manual and non-manual labour. Capital is assumed to be perfectly mobile internationally, and available at a constant price. Other factors are internationally immobile, so their prices adjust to equate demands to endowments. The commodity structure is defined by Nace three-digit industries, with the rest of each economy aggregated into a single perfectly competitive composite, which is tradeable and which we take

Box 2. What is a computable general equilibrium (CGE) model?

CGE modelling typically requires three stages:

1. In the first stage, the modeller needs to choose an appropriate theoretical model that is to underpin the proposed analysis. From the model the basic equations and equilibrium conditions can be derived. For example, in a trade model such as the one in this paper, it is necessary to derive equations for the amount that will be supplied by producers to each market, the quantity demanded of the goods (both consumer demand and intermediate demand), and similarly for factors of production. At equilibrium, demand is equal to supply in both goods and factor markets.
2. In the second stage, real world data are combined and made consistent with the theoretical model. This process is known as calibration. Hence, for a given year we have data on actual trade flows between countries. Calibration of the model involves choosing the parameter values of the model such that, when the equations of the model are solved, the equilibrium quantities demanded and supplied are exactly the same as those given by the actual data on trade flows.
3. Stages 1 and 2 give a base equilibrium of the model. In the third stage, the modeller chooses an experiment, such as reducing trade costs, and then solves the model again. Having changed one of the parameters in the experiment (trade costs), the equilibrium quantities of the model will now be different from the base quantities. The difference between the two is thus the result of the experiment.

as the numeraire. Each of the manufacturing industries is assumed to be imperfectly competitive, with a number of firms producing differentiated products and production being subject to increasing returns to scale. The list of industries used is given in Table 3.

Demand for differentiated products is modelled as a two-stage process, where the demand for a product aggregate depends on a price index for that aggregate, while demand for an individual variety depends on the price of the variety relative to that of the product aggregate. We assume that firms act as quantity competitors in segmented markets. Each firm chooses sales in each country market, taking as constant the sales of all its rivals in each market. Optimization requires the equation of marginal revenue to marginal cost in each market, where the slope of each firm's perceived demand curve depends on the extent of product differentiation, and on the share of the firm in that market. The key feature of the model, shared with the theoretical treatment underlying the econometric model, is that price–cost margins thus depend on firms' market shares, and increased import penetration causes firms to behave more competitively, lowering their price–cost margins.

5.2. Calibration

Whereas the parameters of the econometric model are estimated on the basis of data series spanning several years, the numerical specification of the model is undertaken first by setting some key parameters and variables, notably those describing concentration and returns to scale on the basis of literature estimates, and then calculating the values of remaining parameters and endogenous variables so that the 1991 base year observations support an equilibrium.

The overall utility function is Cobb–Douglas. The price elasticities of demand for individual varieties depend on the elasticities of substitution in the CES aggregators. For final products, we assume that the base dataset represents a long-run equilibrium in which profits are zero. Technology and firm scale imply a relationship between average cost and marginal cost, and, with the assumption of long-run equilibrium, this also gives a relationship between price and marginal cost. This price–cost margin is supported at equilibrium by two considerations: product differentiation and market power stemming from the degree of concentration in the industry and the form of interaction between firms. We assume that the base case is a segmented market Cournot equilibrium. The number of varieties and the elasticity of demand for an individual variety are then chosen so that the price–cost mark-up is consistent with the assumed scale economies. The calibrated elasticities of demand for each industry are reported in column (3) of Table 3. The final stage of calibration involves positioning demand curves so that consumption of products in each country is consistent with the matrix of production and consumption.

5.3. Data

The three-digit level of aggregation is chosen because this seems to be the most appropriate level at which to model interactions between trade policy and market behaviour (and indeed it is the level of aggregation at which we obtain measures of concentration). At the three-digit level, there are 118 manufacturing industries. This number is reduced first by resolving some incompatibilities in the Nace classifications between trade and industrial statistics. Further, because of data limitations in certain countries in small industries, it was necessary to introduce some aggregation. The model works with 64 imperfectly competitive sectors, listed in Table 3.

Trade data were obtained from the COMEXT data bank, and production data from Eurostat industrial databases.[5] Data on concentration, firm numbers and returns to scale are obtained from the study carried out by Davies and Lyons (1996) and from the survey by Pratten (1988). In a model based on imperfect competition, data on concentration are crucial, and compared with the work done by Gasiorek *et al.* (1992), the availability of concentration data from the recent work of Lyons *et al.*, collected on an EU-wide basis and in a way that is more satisfactory than national census data, should considerably improve the reliability of the model. Column (4) of Table 3 lists the average degree of concentration adjusted for import penetration. The figure reported here is a Herfindahl index, the reciprocal of which gives the number of equivalent-sized firms. Column (8) of the table lists the degree of economies of scale in each industry. The percentage figure refers to the increase in costs as a result of a 50% reduction in output from the minimum efficient scale.

Other industry-specific data required include the share of value added in production; the share of each factor in value added; the elasticity of substitution between different factors of production; and the share of final demand in the output of each industry. Columns (5)–(7) give the share of capital, non-manual labour and manual labour respectively for each industry.

5.4. The impact of the Single Market

The econometric analysis distinguished between the direct demand effects of the SMP and the competition effects. In the CGE model, in order to capture the direct demand effects we make use of the econometric estimates. Specifically, we solve for the change in trade costs that generate the changes in shares in apparent consumption as given by the direct demand effect in Table 2. Consistency with the econometric model is ensured by holding the number of firms, factor prices, intermediate input prices, economies of scale and price–cost margins fixed.

[5] At the three-digit level, the production data for any given year are not complete. A variety of procedures have been employed in order to complete the dataset. See Allen *et al.* (1996) for a fuller discussion.

Table 3. Key data characteristics

Nace	Description	Industry characteristics					
			Share in value added				
		Elasticity	Conc.	Capital	Non-manual labour	Manual labour	Returns to scale
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
22	Metals	16.69	0.0984	0.25	0.48	0.27	1.06
23	Mineral extraction	16.69	0.1017	0.45	0.33	0.22	1.06
241:244	Clay, cement, asbestos	6.39	0.0451	0.37	0.45	0.17	1.14
245:248	Stone, glass, ceramics	15.25	0.0479	0.27	0.46	0.27	1.06
251	Basic chemicals	7.08	0.0376	0.38	0.53	0.10	1.15
255	Paint and ink	13.92	0.1428	0.36	0.54	0.10	1.10
256	Ind. and agric. chemicals	7.14	0.0564	0.40	0.50	0.09	1.15
257	Pharmaceuticals	7.02	0.0459	0.40	0.50	0.09	1.15
258	Soap and detergents	10.89	0.0450	0.39	0.52	0.10	1.10
259	Domestic chemicals	7.85	0.1432	0.34	0.55	0.10	1.15
260	Man-made fibres	9.73	0.3860	0.25	0.51	0.24	1.15
311:313	Metal manufacture	13.67	0.0148	0.26	0.43	0.32	1.08
314:315	Metal structures, boilers	14.41	0.0067	0.21	0.55	0.24	1.07
316	Tools and cans	14.55	0.0065	0.30	0.47	0.23	1.07
321	Tractors and agric. machinery	12.53	0.0199	0.15	0.64	0.22	1.07
322	Machine tools	12.07	0.0052	0.19	0.61	0.21	1.07
323	Textile machinery	12.16	0.0113	0.16	0.63	0.21	1.07
324	Food and chemical machinery	12.31	0.0089	0.27	0.55	0.19	1.07
325	Mining and constr. machinery	12.23	0.0129	0.23	0.58	0.20	1.07
326	Transmission equipment	17.18	0.0180	0.20	0.60	0.20	1.05
327	Paper and wood machinery	12.19	0.0118	0.25	0.56	0.19	1.07
328	Other machinery	8.63	0.0191	0.24	0.57	0.19	1.10
330	Computers and office machinery	18.17	0.1671	0.34	0.61	0.05	1.08
341	Insulated wires and cables	5.80	0.0349	0.30	0.56	0.14	1.15
342	Electrical machinery	5.71	0.0103	0.25	0.60	0.15	1.15
343	Electrical equipment	15.78	0.0418	0.25	0.60	0.15	1.03
344	Telecoms and measuring equip.	16.19	0.0289	0.24	0.61	0.15	1.06
345	Radio and TV	16.36	0.0314	0.27	0.59	0.14	1.06
346	Domestic electrical appliances	14.31	0.0997	0.27	0.59	0.14	1.08
347	Electric lighting	12.09	0.2100	0.30	0.56	0.14	1.10
351	Motor vehicles	11.58	0.1001	0.20	0.49	0.31	1.08
352:353	Motor vehicle parts	10.29	0.0339	0.22	0.47	0.30	1.08
361:363,365	Ships, rail stock, cycles	12.52	0.0947	0.17	0.50	0.33	1.08
364	Aerospace	11.48	0.1303	0.19	0.69	0.13	1.08
37	Instruments	6.91	0.0297	0.27	0.57	0.16	1.15
411,420:423	Grains, pasta, bread	8.59	0.0697	0.29	0.50	0.21	1.11
412	Other foods	9.04	0.0064	0.30	0.49	0.21	1.09
413	Meat products	25.48	0.0244	0.39	0.43	0.18	1.04
414	Dairy products	5.76	0.0185	0.40	0.42	0.18	1.15
415	Fruit and vegetable products	10.32	0.0468	0.31	0.49	0.21	1.09
416:419	Fish products	5.89	0.0471	0.44	0.39	0.17	1.15
424:428	Drinks	22.14	0.0778	0.48	0.39	0.13	1.05
429	Tobacco	33.05	0.3055	0.59	0.27	0.14	1.07
43A,B	Wool, cotton, silk, flax	29.72	0.0104	0.24	0.46	0.30	1.03
436	Knitting	28.35	0.0036	0.28	0.41	0.32	1.03
438	Carpets	8.92	0.0333	0.34	0.40	0.26	1.10
439	Misc. textiles	34.34	0.0370	0.34	0.40	0.26	1.03
441	Leather tanning	27.89	0.0048	0.32	0.38	0.30	1.03
442	Leather products	27.93	0.0047	0.25	0.42	0.33	1.03
451	Footwear	28.08	0.0029	0.25	0.36	0.39	1.03
453	Clothing	27.95	0.0027	0.25	0.43	0.32	1.03
455:456	Household textiles, fur	30.35	0.0090	0.19	0.46	0.35	1.03
461:462	Wood boards	17.50	0.0090	0.30	0.39	0.32	1.05

continued

Table 3. ***Continued***

Nace	Description	Industry characteristics					
			Share in value added				
		Elasticity	Conc.	Capital	Non-manual labour	Manual labour	Returns to scale
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
463:465	Other wood	16.97	0.0098	0.23	0.43	0.35	1.05
466	Cork and brushes	17.69	0.0126	0.25	0.41	0.34	1.05
467	Wooden furniture	16.67	0.0033	0.25	0.41	0.34	1.05
471	Paper and pulp	8.54	0.0074	0.40	0.37	0.23	1.10
472	Processed paper	8.55	0.0128	0.31	0.46	0.23	1.10
473	Print and publishing	6.56	0.0145	0.26	0.57	0.18	1.13
481:482	Rubber	23.99	0.0891	0.31	0.45	0.24	1.05
483	Plastics	16.94	0.0070	0.34	0.46	0.21	1.05
491	Jewellery	16.73	0.0059	0.24	0.50	0.26	1.05
492:493	Musical instruments, photos	17.12	0.0164	0.20	0.53	0.27	1.05
494	Toys and sports	17.94	0.0207	0.35	0.43	0.22	1.05

The CGE model operates with 64 sectors and we have detailed estimates of SMP effects for fifteen sectors arising from the econometric analysis. For the remaining sectors, the size of the trade cost reduction is determined by the aggregate econometric results combined with detailed information from both *ex ante* and *ex post* analyses of the SMP as well as from consideration of the actual changes in shares that occurred. The information used includes: estimates by Davies and Lyons (1996) on the extent to which each sector was affected by non-tariff barriers to intra-EU trade and the vulnerability of the sector to other SMP measures; information arising from the study of Buigues *et al.* (1990); analysis of the twenty industries that over the period 1991–4 experienced the largest change in the share of imports from the EU, and from the rest of the world respectively; as well as information from the European Commission's *Single Market Review*.

This combination of both *ex ante* and *ex post* information provides a good picture as to how sectors have been affected. For the sectors that appear to have been substantially affected, but for which we do not have sectoral estimates, we use the weighted average direct demand effect for the fifteen sensitive sectors as reported in the econometric results (Table 2), and for the remaining sectors we use the aggregate manufacturing figures. For details, see Table D1 in Appendix D.

In order to capture the competition effects, we allow for price–cost margins to change in response to the changing market shares. As in the econometric analysis, we report on the changes in shares in apparent consumption for each industry, and examine the extent to which these shares are affected by allowing firms to adjust their price–cost margins. In Table 4 we detail the results for the fifteen sectors for which we have econometric estimates as well as giving aggregate results. Columns (1)–(3) give the direct demand effect of the SMP as determined by the procedure described above. Columns (4)–(6) give the additional impact on these shares where

Table 4. Impact of the SMP by sector (% point changes)

Nace	Direct demand effect			Competition effect			Overall impact		
	Home (1)	EU (2)	ROW (3)	Home (4)	EU (5)	ROW (6)	Home (7)	EU (8)	ROW (9)
245:248	−4.16	1.77	2.40	0.32	−0.20	−0.12	−3.84	1.57	2.28
251	−4.14	2.36	1.78	0.13	−0.08	−0.04	−4.01	2.28	1.74
257	−1.88	0.38	1.49	0.04	−0.02	−0.02	−1.84	0.36	1.47
314:315	−5.29	4.40	0.90	0.02	−0.02	0	−5.27	4.38	0.90
322	−1.85	−2.33	4.18	0	0.01	−0.01	−1.85	−2.32	4.17
324	−7.06	2.69	4.36	0.07	−0.03	−0.03	−6.99	2.66	4.33
325	−1.67	0.97	0.70	0.01	0	−0.01	−1.66	0.97	0.69
330	−6.80	1.98	4.83	0.78	−0.34	−0.43	−6.02	1.64	4.40
344	−2.70	1.69	1.00	0.07	−0.04	−0.04	−2.63	1.65	0.96
345	−11.77	1.90	9.86	0.43	−0.17	−0.25	−11.34	1.73	9.61
351	−4.80	3.61	1.20	0.13	−0.10	−0.03	−4.67	3.51	1.17
364	−15.34	14.54	0.80	0.46	−0.34	−0.13	−14.88	14.20	0.67
424:428	−6.30	5.90	0.40	0.31	−0.29	−0.02	−5.99	5.61	0.38
453	−2.23	−2.54	5.48	0.02	0.01	−0.02	−2.21	−2.53	5.46
Average of 15 sensitive sectors	−6.60	4.36	2.24	0.23	−0.15	−0.08	−6.37	4.21	2.16
Rest of manufacturing	−2.72	0.77	1.94	0.12	−0.06	−0.06	−2.60	0.71	1.88
Total manufacturing	−4.06	2.02	2.04	0.16	−0.09	−0.07	−3.90	1.93	1.97

price–cost margins are allowed to change (competition effect). Columns (7)–(9) give the overall effect on market shares, which is simply calculated by summing the direct effect and the competition effect.

As with the econometric results, the competition effect serves to increase domestic market shares and to counteract some of the loss in market shares arising from the direct effect of the SMP. This occurs because firms typically have larger domestic shares than shares in export markets; the SMP thus has a bigger relative impact on their domestic shares, hence prices go down by more in domestic markets than in their export markets, resulting in an increase in home shares. While the competition effect on domestic market shares is significantly smaller than reported in the econometric analysis, the largest effects are in the same industries that had the largest effects in the econometric analysis. These are Nace 330 (office equipment), Nace 345 (electronic equipment) and Nace 64 (aerospace). The results are clearly related not only to the size of the experiment in each sector, but also to the underlying characteristics of the industries. For example, the results are closely correlated with the degree of concentration in the industry, which is also generally true of the econometric results except for one industry (Nace 351). These results are also reflected in the aggregate numbers, which not surprisingly report a larger direct effect for the sensitive sectors, and a correspondingly larger competition effect.

An important difference between econometric and CGE analyses is that the former capture any factors that are not included (controlled for), while the latter

include only effects that are directly modelled. As the econometric results show, the SMP has clearly had a pro-competitive impact and in principle this could arise from the impact of the SMP on the *intensity* of competition or on the *nature* of competitive interaction. The change in intensity arises in the first instance because of the direct change in market shares, but could also reflect changes in market shares due to changes in firm numbers. Changes in the nature of competitive interaction imply that firms adjust the way in which they set their prices, and hence their price–cost margins. These possible effects are implicitly captured in the econometric results, but are not distinguished. In contrast, in the CGE model, to capture such effects we need to include them explicitly in the analysis. For example, in the previous experiment we allowed price–cost margins to change, but held firm numbers constant. Yet changes in firm numbers would also change the intensity of competition that may be being captured in the econometric analysis. We have also assumed that firms compete in segmented markets as opposed to allowing for any changes in competitive interaction.

In Table 5 we report on three further experiments that allow for such changes. The table now reports the results for all of the industries, and in the table we now focus only on the changes in *home* shares in apparent consumption. Columns (1) and (2) of Table 5 repeat columns (1) (direct demand effect) and (4) (competition effect) of Table 4. In column (3) we change the nature of competitive interaction and allow for firms to adjust their price–cost margins, but in the face of integrated markets. We hold all else (firm numbers, input costs, scale) constant, so direct comparison should be made with column (2). In columns (4) and (5) we allow for the full interaction of general equilibrium effects through allowing firm numbers and wages to be flexible for both segmented and integrated markets.[6] Columns (2)–(5) report the change in home market share additional to the change induced by the direct demand effect of the SMP. The total effect in each case, therefore, would be given by the sum of the direct effect and the additional effect.

Looking first at the short-run integrated markets experiment, it can be seen that, as with segmented markets, in all cases home market shares increase. However, the orders of magnitude are significantly larger and are much more in line with the econometric results reported earlier. Indeed, in ten cases the initial decline in home market shares arising from the direct demand effect is overturned as firms adjust their price–cost margins. The sectors experiencing the largest changes are Nace 22 (metals), Nace 481:482 (rubber) and Nace 255 (paint and ink). These changes occur because of the much greater pro-competitive impact of the SMP under the integrated markets assumption. With an integrated market pricing rule, it is firms' market share across the whole EU that determines their price–cost mark-ups in

[6] For two industries (Nace 260 and Nace 429) and in certain (smaller) countries, the long-run integrated market experiment resulted in completely implausible and unstable numbers. For this experiment, therefore, for these industries the experiment was set at zero.

Table 5. Exploring further the effects of the SMP (% point changes)

Nace	Direct effect	Competition effect		Long run	
		Segmented markets	Integrated markets	Segmented markets	Integrated markets
	(1)	(2)	(3)	(4)	(5)
22	−2.18	0.38	8.72	0.29	8.18
23	−2.20	0.16	3.13	0.04	2.44
241:244	−2.13	0.01	0.41	−0.04	0.26
245:248	−4.16	0.32	4.08	−0.06	3.05
251	−4.14	0.13	1.76	−0.32	0.25
255	−5.46	0.69	5.93	0.25	3.77
256	−5.14	0.24	2.95	−0.43	0.31
257	−1.88	0.04	1.31	−0.08	0.51
258	−5.40	0.23	1.94	−0.19	0.24
259	−2.11	0.15	4.00	−0.09	2.19
260	−2.18	0.17	1.27	0.03	
311:313	−2.20	0.02	0.34	−0.01	−19.49
314:315	−5.29	0.02	0.16	−0.04	2.97
316	−2.18	0.01	0.37	0.02	−7.44
321	−2.13	0.03	0.75	−0.12	0.97
322	−1.85	0	0.10	−0.20	−1.02
323	−2.00	0.01	0.35	−0.08	0.46
324	−7.06	0.07	0.39	−0.39	5.46
325	−1.67	0.01	0.34	−0.01	−5.22
326	−2.13	0.03	0.52	0.06	0.07
327	−2.09	0.02	0.34	0.02	1.29
328	−2.11	0.02	0.44	−0.13	0.15
330	−6.80	0.78	3.97	−0.17	0.25
341	−5.14	0.02	0.15	−1.53	2.20
342	−4.74	0.02	0.15	−0.86	−10.56
343	−5.45	0.24	2.41	0.03	−6.90
344	−2.70	0.07	1.30	0.11	−0.76
345	−11.77	0.43	1.52	−2.25	9.50
346	−2.10	0.24	5.60	−0.07	−10.20
347	−2.13	0.26	5.83	−0.01	1.42
351	−4.80	0.13	1.62	−0.04	4.66
352:353	−2.13	0.02	0.43	−0.06	−1.97
361:363,365	−5.44	0.20	1.66	−0.82	0.17
364	−15.34	0.46	1.81	−0.26	7.48
37	−4.94	0.05	0.48	−0.28	−3.94
411,420:423	−2.21	0.06	1.26	−0.23	−8.30
412	−2.15	0	0.07	−0.43	−2.67
413	−2.20	0.06	0.87	−0.09	−2.63
414	−2.21	0.01	0.19	−0.83	−1.93
415	−2.12	0.06	1.32	−0.59	−8.37
416:419	−5.18	0.02	0.22	−0.11	4.63
424:428	−6.30	0.31	1.93	−0.39	−51.00
429	−2.20	0.80	2.54	0.93	
43A	−3.00	0.09	1.54	0.08	−11.44
436	−2.23	0.12	1.37	0.02	−2.47
438	−2.12	0.04	1.01	−0.74	0.50
439	−2.19	0.19	3.87	0.20	0.88
441	−2.98	0.02	0.25	0.25	−2.13
442	−2.23	0.02	0.41	0.07	0.30

continued

Table 5. ***Continued***

Nace	Direct effect	Competition effect		Long run	
		Segmented markets	Integrated markets	Segmented markets	Integrated markets
	(1)	(2)	(3)	(4)	(5)
451	−2.23	0.03	0.39	−0.43	−1.81
453	−2.96	0.02	0.30	−0.56	−1.38
455:456	−2.21	0.10	1.57	−0.07	−3.67
461:462	−2.21	0.06	0.98	−0.12	17.57
463:465	−2.20	0.01	0.21	−0.05	5.17
466	−2.22	0.09	1.45	−0.56	−3.84
467	−2.20	0.01	0.08	−0.18	−0.59
471	−2.01	0.03	0.58	−0.32	−5.90
472	−2.16	0.02	0.49	−0.09	−0.10
473	−2.18	0.01	0.15	−0.04	1.85
481:482	−5.50	1.03	10.60	0.61	0.09
483	−2.19	0.03	0.58	−0.04	−22.00
491	−2.17	0.02	0.31	0.08	8.29
492:493	−2.20	0.02	0.58	−0.02	7.51
494	−2.03	0.12	2.20	0.08	1.03
Average of 15 sensitive sectors	−6.60	0.23	1.68	−0.27	−1.20
Rest of manufacturing	−2.72	0.12	1.62	−0.10	−7.00
Total manufacturing	−4.06	0.16	1.63	−0.16	−4.49

each market. With segmented markets, the large market shares of home firms allow them to set high price–cost margins, which encourage intra-EU imports (as in the reciprocal dumping model of trade). Market integration removes the effect of home market bias on firms' behaviour, and firms cut price–cost mark-ups particularly in their home market, thus increasing their market shares and driving down imports. Although market integration implies a significant change in market conduct, there is nevertheless a close correlation in the segmented and integrated markets results. This can be clearly seen by looking at Figure 4.

Allowing for the full interplay of general equilibrium interactions has two implications. First, allowing for firm numbers to change implies changes in concentration and thus changes in the intensity of competition. The direct effect of the SMP is to reduce home shares while increasing export shares. As most firms have larger domestic as opposed to export sales, this implies a decline in profits, and thence the exit of firms. The consequent increase in each remaining firm's market share tends to increase price–cost mark-ups; those mark-ups and hence prices will tend to rise by more in the home market than in export markets. Allowing for firm numbers to change could therefore result in a further decline in the home market in addition to the direct demand effect. The second implication is that allowing for changes in factor prices will tend to dampen down the effects previously seen.

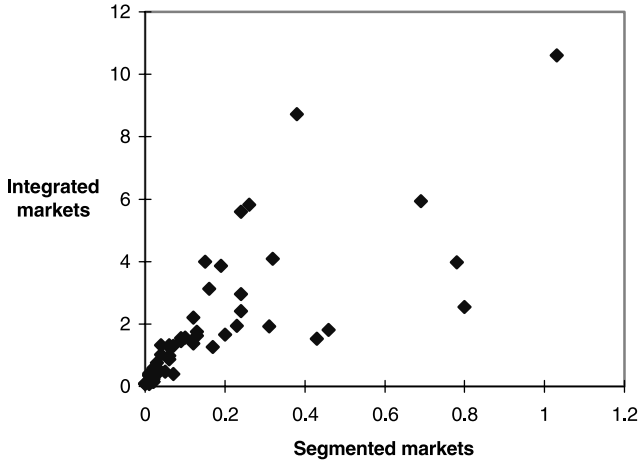


Figure 4. Short run – correlation of integrated and segmented markets experiments

Hence, where only price–cost margins were allowed to adjust, firms were able to recapture some of the lost market share. In the long run this is no longer necessarily the case. With segmented markets in eighteen industries, there is attenuation of the initial loss of market shares arising from the direct demand effect. For the remaining industries, there is a further decline, and in particular for Nace 345 (radio and TV) and Nace 341 (insulated wires and cables). In comparison, with integrated markets 35 industries see the attenuation of the initial loss of home shares, and in seven cases this initial loss is more than reversed. This again reflects the greater pro-competitive impact of this experiment. However, it should also be noted that again the magnitudes of the changes, both positive and negative, are much larger with integrated markets. Hence some sectors experience a dramatic further loss in home shares, with seven sectors experiencing a further decline in shares of more than ten percentage points. With integrated markets not only are the magnitudes significantly higher, but also the pattern of results across industries is quite different in comparison to the segmented market experiment. This can be seen in Figure 5.

These results serve to reinforce both the *ex ante* literature on the SMP (e.g., Smith and Venables, 1988) and the theoretical literature, all indicating that a move to integrated markets results in a much greater pro-competitive impact. The results also show that the greater the pro-competitive impact, the greater the degree of restructuring by sectors, and that the pattern of restructuring depends more on the industry characteristics as opposed to the size of the initial trade barrier removal as given by the direct demand effect. It is also important to note that the assumption that the SMP goes all the way to a single market is probably too bold. On the other hand, assuming that markets remain segmented is in turn perhaps too cautious (Sleuwagen and Yamawaki, 1988); and this is also supported by the comparison between the econometric and CGE results. The integrated markets assumption might plausibly then represent an upper bound.

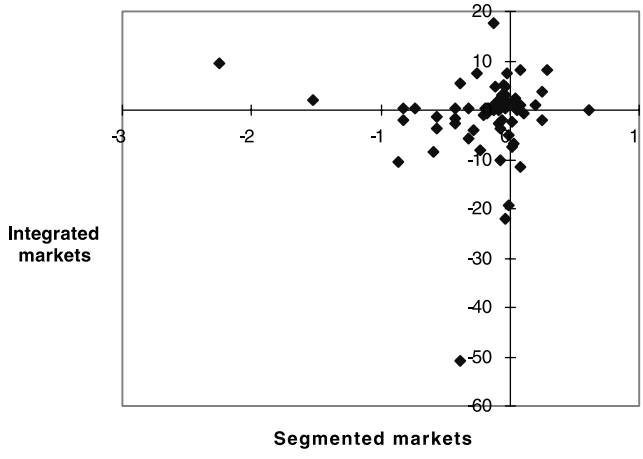


Figure 5. Long run – correlation of integrated and segmented markets experiments

5.5. The impact of the Single Market by country

In the preceding analysis we examined the effects of the SMP by industries. One of the advantages of CGE modelling is the possibility of examining the effects on each country, as well as looking at the welfare consequences of the reductions in trade barriers. Table 6 presents the results of the same experiments as in Table 5, but disaggregated by country.

From the table it can be seen that direct demand impact is on the whole greatest for the larger economies of the EU, with the smallest impact on home shares being experienced by Portugal, and the greatest by France. It is also notable that for

Table 6. Exploring further the effects of the SMP – results by country (% point changes)

Country	Direct effect	Competition effect		Long run	
		Segmented markets	Integrated markets	Segmented markets	Integrated markets
France	−5.06	0.11	0.97	−0.38	−0.79
Germany	−4.24	0.08	0.59	−0.16	−3.42
Italy	−3.35	0.13	1.42	−0.28	−2.00
UK	−3.88	0.12	1.27	−0.30	−4.34
Netherlands	−3.90	0.27	3.14	0.04	0.39
Belgium–Luxembourg	−4.20	0.29	2.42	0.22	0.52
Denmark	−3.96	0.55	5.81	−0.40	0.42
Ireland	−3.09	0.48	5.34	0.10	2.88
Greece	−3.30	1.25	10.57	0.84	12.55
Spain	−2.84	0.25	2.23	−0.24	−4.64
Portugal	−2.19	0.28	5.07	0.92	5.79

France, Germany, Italy, the UK and Spain, the competition effect for both segmented and integrated markets tends to be significantly smaller than for the remaining countries. These results would appear to suggest that, although the direct impact of the SMP may be smaller for the smaller economies (and this arises because their base import penetration ratios are lower), there may be greater restructuring taking place.

As before, much larger results are obtained when we allow for integrated markets (short run). For all countries the decline in home shares experienced as a result of the direct demand effect is reversed, and for several of the smaller economies (Denmark, Ireland, Greece and Portugal) it is overturned. It is notable that the integrated market effect is again much larger for the smaller economies. This suggests higher domestic market shares for home firms prior to integration, and hence a much greater impact as firms switch to an integrated markets pricing rule.

As discussed before, allowing for the free entry and exit of firms, as well as flexible factor prices, changes the results in important ways. Hence with both segmented and integrated markets the larger economies experience a further decline in their domestic market share, while the smaller economies in general reduce the initial loss arising from the direct demand impact. These impacts are significantly larger in the integrated markets case.

Overall these results reinforce the econometric results concerning the pro-competitive impact of the SMP. Under segmented markets, allowing firms to adjust their price–cost mark-ups enables them to recapture some of their market share. However, these results are of a smaller order of magnitude than seen in the econometric results. This suggests that the effects of the SMP on competition in Europe should perhaps not be seen simply as increasing the intensity of competition as a result of the decline in barriers. One possible explanation is that, as well as changing the intensity of competition, there may be a change in the *nature* of competition. We have modelled this change as a switch to integrated markets, and with that assumption the results are much more similar to those derived econometrically.

There may, of course, be other factors that we are not capturing. For example, on the basis of the switching costs literature, it can be shown that the higher the firm's existing market share, the greater is the propensity to reduce price–cost margins in order to maintain that share in the face of a decline in trade costs (Allen, 1994). Equally, we have assumed that all firms act as quantity competitors in either segmented or integrated markets. Clearly, the reality is likely to be more complex, with some firms being characterized by other forms of competition such as price competition; or integration inducing a change from quantity competition towards price competition (Klemperer and Meyer, 1989). Finally, the extent to which integrated markets are being achieved is likely to vary across sectors.

5.6. Welfare

In this section we discuss some of the overall welfare implications of the various experiments reported above. Turning first to the overall welfare numbers, these are reported in Table 7. Column (1) of the table reports the welfare impact of the direct demand effect of the SMP.[7] The subsequent columns report the overall welfare effects, but with the progressive relaxation of the assumptions as outlined previously. Columns (6)–(10) detail the changes in welfare, but this time as a percentage of base value added in the manufacturing sector.

Not surprisingly, the welfare effects are positive and are on the whole increasing as one moves through the different experiments. For the first two experiments for Greece, there is a small welfare loss. This is a result of holding certain variables constant in these experiments (scale, firm numbers, intermediate input costs), which thus allows for negative profits that can generate a welfare loss. This can be seen most starkly in the short-run integrated markets experiment, where all countries experience a welfare loss. This is entirely accounted for by the decline in profits that firms experience as a result of the strong pro-competitive impact of the experiment.

In the long run, for both segmented and integrated markets the welfare effects are positive and in general higher for the smaller economies, and this is particularly so with integrated markets. Once again this can be understood with reference to the greater pro-competitive impact of the SMP for the small economies. These results indicate that the long-run welfare effects are potentially very large – indeed, in certain cases (particularly with integrated markets) the results are probably implausibly large. We report them here for the sake of completeness, but they need to be treated with caution. As discussed above, this occurs because the switch to an integrated markets pricing rule leads to dramatic changes in firms' conduct. Nevertheless, it is worth emphasizing that the experiments in this paper apply only to manufacturing, which comprises approximately 30% of EU countries' GDP. In that context, the changes in welfare are substantial.

6. CONCLUSIONS

The analysis in this paper has been based on two complementary approaches – econometric modelling and computable general equilibrium analysis. The main conclusions from the econometric analysis are that the Single Market Programme has indeed had a substantial pro-competitive effect in European markets, and has led to significant reductions in price–cost margins. The reduction in price–cost margins has in turn been greater in domestic markets, leading to a smaller decline in domestic shares than would otherwise have been the case.

[7] The welfare numbers reported here are the compensating variation.

Table 7. Welfare effects of the SMP

	As a % of base GDP					As a% of base manufacturing value added				
	Direct effect	Competition effect		Long run		Direct effect	Competition effect		Long run	
		Segmented markets	Integrated markets	Segmented markets	Integrated markets		Segmented markets	Integrated markets	Segmented markets	Integrated markets
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
France	0.20	0.18	−0.15	0.51	0.67	0.76	0.69	−0.56	1.94	2.73
Germany	0.26	0.24	−0.01	0.56	0.77	0.80	0.75	−0.04	1.74	2.56
Italy	0.10	0.08	−0.39	0.43	0.75	0.34	0.26	−1.30	1.41	2.63
UK	0.18	0.16	−0.35	0.58	1.11	0.65	0.56	−1.24	2.03	4.23
Netherlands	0.28	0.25	−0.65	0.75	1.72	1.08	0.96	−2.52	2.91	7.17
Belgium–Luxembourg	0.32	0.28	−0.99	0.94	1.88	1.21	1.03	−3.71	3.54	7.44
Denmark	0.08	0.08	−0.87	0.54	2.76	0.45	0.45	−4.63	2.87	15.17
Ireland	0.14	0.19	−3.54	0.93	9.54	0.45	0.64	−11.80	3.10	32.34
Greece	−0.40	−0.10	−0.13	1.21	15.46	−2.64	−0.67	−0.89	8.05	104.00
Spain	0.12	0.08	−1.16	0.62	1.70	0.39	0.28	−3.83	2.06	5.99
Portugal	0.78	0.75	−3.66	3.01	17.63	2.36	2.26	−11.00	9.05	53.08

These conclusions are strongly supported by the computable general equilibrium analysis. The econometric and secondary evidence used suggests that the SMP has had a significant direct impact on a wide range of sectors. This direct impact has in turn resulted in a significant pro-competitive impact, causing firms in the first instance to adjust their price–cost margins. The extent of this adjustment, as well as of any subsequent restructuring, depends strongly on the assumptions that are made regarding the nature of competitive interaction. The comparison between the econometric and the CGE results suggests that the impact of the SMP is to change not simply the intensity of competition, but possibly also the nature of competitive behaviour in Europe. This indicates that the European Union might indeed have indeed moved some distance away from a world of segmented markets.

Perhaps unsurprisingly, the evidence suggests that the SMP has been strongly welfare increasing for all economies, but that there is considerable variance in the distribution of these welfare gains across the EU economies. The larger welfare gains are experienced by the smaller economies, and this occurs because of the greater pro-competitive impact of the reduction in barriers. Nevertheless these larger welfare gains in turn imply a greater degree of restructuring. Finally, it is clear that the welfare gains rise the greater the degree of integration, and there is a substantial difference in welfare gain in comparing the segmented and integrated markets results. In both cases, however, the welfare gains can be seen to be quite substantial.

Discussion

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The Single Market Programme raised high hopes of a rejuvenation of the European economy when it was launched in 1985. Europe had been slipping behind in economic growth, and trade with non-European countries was rising faster than with European countries. The Single Market was going to increase trade between member states, increase the degree of competition, make European firms more competitive on world markets, and increase the rate of growth.

Although 1 January 1993 should be seen as the starting date of the Single Market, only about 55% of the 219 measures had been implemented by all fifteen member states by September 1996 (European Commission, 1996a). The European Commission started to evaluate the effects of the Single Market as early as 1995. That is surely too early to see the long-term effects, which were expected to involve substantial restructuring of ownership, production and trade. The results of a large number of studies of the effects of the Single Market by the Commission and independent experts were published in European Commission (1996a). It should not come as a surprise that the verdict was quite positive.

The present paper follows in the wake of the Commission's evaluation. First, it attempts to estimate the size of the reduction of trade costs in fifteen manufacturing industries thought to be especially affected by the Single Market, using econometric methods. (An earlier version is part of the evaluation by the Commission.) Second, it attempts to estimate the welfare effects by running simulations with a computable general equilibrium model. This is potentially a very valuable complement to the Commission's work, since welfare is what one is interested in ultimately.

Descriptive statistics

Before turning to more sophisticated econometric and simulation techniques, the paper presents some descriptive statistics for an indication of the impact of the Single Market. Since the Single Market eliminated trade costs and market access restrictions between member states of the EU, one should expect it to affect trade volumes. In particular, intra-EU trade should increase (trade creation) and non-EU imports should be diverted to intra-EU imports (trade diversion). The Single Market made it less costly also for non-EU producers to supply the EU, but probably not as much as for EU producers.

Figure 1 shows in what parts manufacturing demand in EU countries was supplied by domestic production, imports from other EU countries and imports from non-EU countries between 1980 and 1994. It is clear that the share of domestic production was falling throughout the period, and that both EU and non-EU imports have been increasing steadily. The noteworthy exception is EU imports since 1992. It is only when EU imports are adjusted to account for assumed underreporting of intra-European trade because of changes in data collection that the EU share is seen as rising after 1992.

It should be noted that Figure 1 shows no marked change in the time series since the Single Market was implemented (or at any other time), and that EU and non-EU imports have done equally well. This is also the impression from the disaggregated, industry-level time series in Appendix A. Thus, the descriptive statistics do not suggest reasons to believe that the Single Market had important effects on trade.

At this point, one should stop to ask why trade effects are important, and if the Single Market could produce welfare gains even if shares of apparent consumption seem to be unaffected. Theoretically, the Single Market can be expected to affect trade in several ways. First, it should contribute to a further exploitation of comparative advantage due to technology or factor endowment differences across countries, and thus to increased trade. Second, by eliminating trade costs and market access restrictions, it should lead to increased competition. This should in turn result in smaller price–cost margins, greater returns to scale and greater cross-market penetration (not based on comparative advantage) – i.e., to more trade. Greater exploitation of comparative advantage, smaller profit margins and greater scale economies all serve to improve economic welfare. (Welfare is also improved by

the increased variety in inputs and in final consumer goods and services, and by the reduction of real trade costs.) Thus, for the Single Market to be beneficial, it should change trade volumes and patterns. The reduction in trade barriers and increased market access make markets more *contestable* and can lead to more competition without necessarily causing trade to increase; domestic firms lower their profit margins to prevent foreign firms entering their market.

Econometrics

It is, of course, entirely possible that the descriptive statistics are the net result of many different influences, such that the effects of the Single Market have been neutralized by other factors. In order to control for other factors and estimate the pure effects of the Single Market on shares of apparent consumption, econometric techniques are employed. Demand and price functions have been estimated for domestic production, EU imports and non-EU imports respectively in fifteen manufacturing industries in the four largest EU countries, comprising about one-third of total manufacturing value added. Demand is modelled as determined by relative prices and real income, and supply by various costs, competitors' prices and the business cycle.

The estimation is done on annual data for the period 1976–94. In view of the fact that the Single Market started as late as 1993, one must question the exclusion of years later than 1994. The ability of the econometrics to establish significant effects of the Single Market should be strongly improved by the inclusion of data for at least 1995 and 1996 (which should be available by now).

To capture the Single Market effects, dummy variables were included for 1992–4. The problem with this is that the dummy variables will capture all factors that made demand and price setting shift in 1992–4. In other words, one cannot be certain if and to what extent the estimates capture Single Market effects. A further problem is that a dummy variable was included for 1992, although the Single Market was effectively launched in 1993. This aggravates the problem of interpreting the estimates for the dummy variables.

Ideally, an attempt should have been made to determine if and when there are structural breaks in the time series for consumption shares. As it is, one does not know if there were significant shifts in the demand and pricing functions prior to 1991. If these took place, the interpretation of the shifts in 1992–4 becomes even harder.

All of this makes the econometric estimates of the Single Market effects presented in Table 2 quite uncertain. That aside, it is noteworthy that the estimated overall impact of the Single Market for the average of the fifteen 'sensitive' sectors is as large for the non-EU share of apparent consumption as it is for the EU share. This is consistent with the descriptive statistics. Moreover, for the remaining two-thirds of manufacturing, the Single Market impacted *negatively* on the EU share, while the

non-EU share increased. It is hard to find reasons why the Single Market should benefit non-EU producers more than EU producers. One may speculate that non-European producers, for which the cost of supplying the whole EU market has decreased, tend to be larger and see the EU as one market, while EU producers tend to be smaller and have a more limited market scope.

The CGE experiments

Econometric estimation has the advantage of being based on actual data and – in this case – on simple and robust economic theory. It has the disadvantage that it does not capture the complicated interactions in product and factor markets throughout the economy. Furthermore, it cannot be used to answer the fundamental question: namely, what the welfare effects of the Single Market are. Numerical models of the whole economy capture the market interactions and can be used to isolate various effects of the Single Market and to estimate its overall welfare effect.

But all computable general equilibrium models share common weaknesses. First, the structure of the models is not self-evident and will, of course, influence the results. For example, in the present model, individual industries are modelled as being oligopolistic. If they instead were modelled as characterized by monopolistic competition, the pricing functions would tend to become flatter, which in turn would tend to make for greater effects for a given change in policy. It is not clear which market structure is more realistic. Also, one must make certain that the model is stable – i.e., that it does not generate unrealistically large effects for small changes in exogeneous variables. The present model is apparently unstable when it is made fully flexible: that is, when prices, quantities and the number of firms are made endogenous. Second, the results are sensitive to the values assigned to various economic parameters, such as the sensitivity of demand to price or the ease with which one factor of production can be substituted for another. The present model has the advantage that it can use as the policy change the estimated reductions in trade costs from the econometric part of the paper. At the same time, estimates exist for only fifteen of 64 manufacturing industries that are included in the model. The rest is guesswork. Another data problem is that the starting number of product varieties and the sensitivity of demand were calculated on the basis of parameters for scale economies in the various industries. The parameter values were in part taken from a study with engineering estimates from the 1960s (Pratten, 1988). One wonders to what extent these data are still valid.

In view of these and other shortcomings, the numbers generated by the various experiments must be looked at with scepticism. Again, it appears that non-EU producers have gained more than EU producers in terms of market shares (see Tables 5 and 6), which is hard to explain. A peculiar result is the very large welfare gains for some of the smaller countries (see Table 7). The gains are stated relative to

total income and are unbelievable relative to manufacturing, which makes up roughly 30% of GDP. Such numbers generate distrust in the rest of the results.

Final remarks

My comments may seem quite negative: the descriptive statistics do not indicate any significant effects of the Single Market on trade patterns, the econometric evidence is hard to interpret, and the results of the CGE experiments should be regarded with considerable scepticism. This is not to say, however, that the Single Market has had insignificant effects. The point is rather that our methods of estimating some of the effects are deficient, and that insufficient time has elapsed since the Single Market came into existence.

Peter Birch Sørensen

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This paper makes a bold attempt to evaluate the effects of the Single Market Programme (SMP) on market shares and country welfare in the EU. It does so by combining econometric evidence with simulations of a computable general equilibrium model which allows for imperfect competition in product markets and encompasses a large number of industrial sectors. The authors' econometric methodology enables them to separate the direct effect on market shares generated by the fall in trade costs from the indirect effect arising as firms change their price–cost margins in response to increased competitive pressure.

In the CGE analysis of the effects of the SMP, the change in trade costs is calibrated so as to generate a direct effect on domestic market shares corresponding to the direct impact on market shares estimated in the econometric analysis. The CGE model then allows the authors gradually to activate a number of behavioural and general equilibrium responses to the direct change in market shares to illustrate the various indirect competition effects predicted by the model. These effects are finally compared to the competition effects on market shares estimated in the econometric part of the analysis. The CGE model tends to generate smaller competition effects than those estimated econometrically, unless it is assumed that the SMP has led to a switch from segmented markets with price differentiation across countries to integrated markets where each firm charges the same price for its product throughout the EU. On the basis of their CGE model, the authors also conclude that the SMP has generated substantial welfare gains, particularly in the smaller EU economies, and especially if one assumes that the SMP has induced a switch to integrated markets.

The paper represents an impressive research effort, and it is quite creative in the way in which it separates the direct demand effects of changing trade costs from the indirect competition effects, and in combining econometric and CGE analysis. The authors have thus made a valuable contribution to our understanding of the effects

of the SMP. Yet the paper leaves a number of questions unanswered, or at least underresearched.

The first question arises from Figure 1 in the paper. This figure illustrates a long-term trend towards increasing import penetration ratios in EU manufacturing, but it is very hard to spot any significant effect of the SMP in the late 1980s and early 1990s. If the SMP really generated the large welfare gains estimated by the authors, one would have expected to be able to observe a noticeable impact on the import penetration ratio with the naked eye.

The second question arises from studying Table 2, which summarizes the findings from the econometric analysis. According to the average figures in the three bottom rows of the table, the domestic market shares of firms from the rest of the EU and from the rest of the world (ROW) have increased by roughly the same amount in the fifteen sectors that were most sensitive to the SMP, but for the manufacturing sector as a whole, the market share of ROW has increased considerably more than the share of other EU producers. This is difficult to understand. If anything, one would have expected the SMP to provide a greater stimulus to intra-EU trade than to trade with the rest of the world. Of course, it is quite conceivable that the harmonization or mutual recognition of technical standards within the EU facilitates the market access of ROW and EU producers to the same extent. While this may explain the econometric findings for the SMP-sensitive sectors, it still does not explain why the SMP apparently generated so much more import penetration from ROW in the rest of the manufacturing sector. Moreover, some elements of the SMP must surely have generated a larger fall in intra-EU trade barriers than the fall in barriers against the rest of the world. For example, while public procurement has in principle been liberalized within the EU, we have not seen a similar liberalization of public procurement *vis-à-vis* the rest of the world. Similarly, the reduction of some of the physical and fiscal trade barriers has not been extended to trade with ROW. The large positive impact of SMP on ROW market shares therefore remains a mystery, and it is disappointing that the authors make no effort to explain this surprising finding.

I have so far accepted the authors' econometric estimates at face value. However, I am concerned about their procedure of capturing the SMP effect by introducing a dummy variable for the short period of 1992–4. Surely, we know that the implementation of the SMP has extended over a much longer period, starting in the late 1980s, and that the process has not yet been quite completed. It may be true that the implementation of the various SMP directives accelerated somewhat in the 1992–4 period, but surely there must have been an SMP effect in earlier as well as later years. For example, there is econometric evidence that the SMP started to put downward pressure on price–cost margins right from 1987 (see European Commission, 1996a). The authors report that the use of dummies for longer time periods implies stronger effects of the SMP. Why were these estimates not used instead of postulating that all of the SMP effects materialized in 1992–4?

The use of the dummy variable technique to capture the effects of the SMP is a natural procedure only if one has controlled properly for other factors influencing the evolution of market shares. In principle, the authors' dummy variable could pick up any influences on market shares other than the impact of total spending and relative prices which are used as explanatory variables. For instance, if for some reason the general process of globalization accelerated in the early 1990s, this effect would be attributed to the SMP dummy in the authors' analysis. Indeed, if the estimated coefficient of the authors' dummy does to some extent capture the more general forces of globalization, this would explain why the SMP was estimated to have such surprisingly large effects on the market shares of firms from the rest of the world.

Another concern about the econometric analysis is that the estimated competition effects for the rest of manufacturing in the second row from the bottom of Table 2 are hard to understand. One would think that domestic producers would have reacted to the initial loss of market share by cutting their mark-ups so as to recoup some of their lost market share. According to the estimates, however, the opposite seems to have happened.

The authors offer the hypothesis that the SMP may have changed the nature of competition by inducing a switch towards integrated markets with no cross-country price differentiation. As already noted, this assumption would help to bridge the gap between the estimated competition effects in the econometric analysis and in the CGE analysis. If many sectors have indeed switched to integrated market pricing, this would also help to resolve the apparent inconsistency between the claim that the SMP has generated large welfare gains and the fact that aggregate market shares have not changed very much (Figure 1). The point is that, if domestic firms start to consider the entire EU as a single market that is fully integrated with the domestic market, their perceived market share will often fall dramatically, since domestic firms frequently have a dominant initial position in the domestic market. Domestic firms will thus tend to cut their mark-ups by much more if they perceive a switch to an integrated market, and thereby will recoup most of their initial loss of domestic market share. The effects of the SMP will then show up mainly as a fall in profit margins, bringing prices closer in line with marginal costs, rather than as a significant rise in import penetration.

While it may be tempting to accept this story, the snag is that the welfare effects of the SMP predicted by the authors' CGE model seem implausibly large in the scenario with a switch to integrated markets. Welfare effects amounting to 10–20% of GDP for the smaller economies are simply to good to be true (see Table 7), and the authors do indeed conclude that the assumption of a full switch to integrated markets is probably too strong.

Like other CGE models, the model applied by the authors neglects the short- and medium-term adjustment costs that follow from substantial industrial restructuring. These costs are real, and in fact they are often the reason why policy-makers and

the public shy away from measures of deregulation and liberalization that promise substantial long-term efficiency gains. By abstracting from adjustment costs, the present analysis tends to overestimate the welfare gains from the SMP. On the other hand, by assuming perfect competition in the labour market, the model neglects the fact that the fall in profit margins induced by the SMP will tend to reduce structural unemployment. Welfare will thus improve not only because of smaller distortions in product markets, but also because of lower involuntary unemployment. The authors' model is quite sophisticated in its description of product market imperfections. In future work, CGE model-builders should also pay attention to labour market imperfections and should try to model adjustment costs of industrial restructuring, difficult though this may be.

General discussion

Philippe Aghion criticized the paper's sole focus on homogeneous firms. In a model with entry in the first stage and competition in the second stage, it is well known that *ex post* competition drives out entry. It seems that there is some trade-off between *ex ante* and *ex post* competition. However, this trade-off quickly disappears when we allow for heterogeneity among firms. *Ex post* competition helps good firms to increase their market shares at the expense of bad firms. Hence, even if the market share of domestic firms does not vary much, competition may have a big effect in terms of composition. There is no composition effect in the paper and it therefore underestimates the welfare gains of competition. In addition, competition stimulates restructuring, thus making more of the domestic firms competitive.

Paul Klemperer noted that the effects of the Single Market Programme (SMP) may have changed over time. For instance, opening the market may initially lead to more intensive competition, which is followed by a period of consolidation. Carmen Matutes argued that it is not so surprising that the SMP should benefit the rest of the world, as outsiders will find it more attractive to enter the EU market. It is, however, surprising that the competition effect does not always diminish this entry effect. Georges de Ménil noted that the SMP was also created to unify factor markets. If that is thought to have happened, there was presumably some effect on the trade. There is some element of a missing story in that all calculations assume that labour is immobile.

Jonathan Eaton wondered whether the SMP is still mainly concerned with final consumers. A few years ago, there was the case of a French automobile maker who price-discriminated among retailers in France and Belgium. The court ruled that the Single Market applies only to final consumers, thus allowing producers to price-discriminate among retailers. Jakob de Haan supported the use of dummy variables to differentiate between the periods before and after 1992. If the same dummy variable is applied to all sectors, the important feature that sectors are affected

differently by the SMP will be missed. Samuel Kortum suggested including countries other than those in the EU, so as to exclude pure time effects and to isolate the incremental effect of the SMP on EU countries. Michael Waterson wanted to know which unit of analysis is used in measuring market shares and what happens to firms operating in several countries.

Samuel Bentolila noted that there is a structural change in the data in the period 1992–4, where there is a switch from border data to firm reporting. He suggested using firm data for the entire period. In addition, Bentolila felt that there is a need for more econometric detail. In particular, we want to know which worldwide variables are controlled for – the recession in the early 1990s, for example, could have affected the numbers. The paper also does not say how it treats East Germany. East Germany's transfer from the rest of the world to the EU could have had a significant effect on the trade figures. Finally, the paper implies that the EU share in trade with Portugal falls as a consequence of the SMP. This is very unlikely, as Portugal was a latecomer to the EU. And, in this period, there was a lot of trade diversion to the EU from the rest of the world.

APPENDIX A. CHANGES IN SHARES IN APPARENT CONSUMPTION BY NACE TWO-DIGIT INDUSTRIES

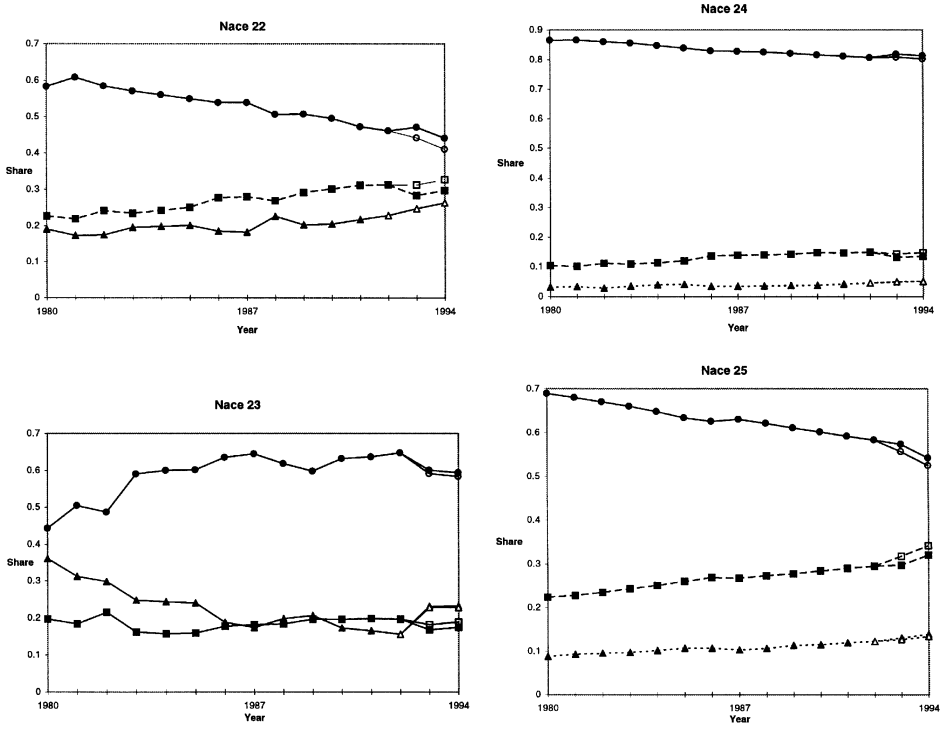


Figure A1. ***Continued***

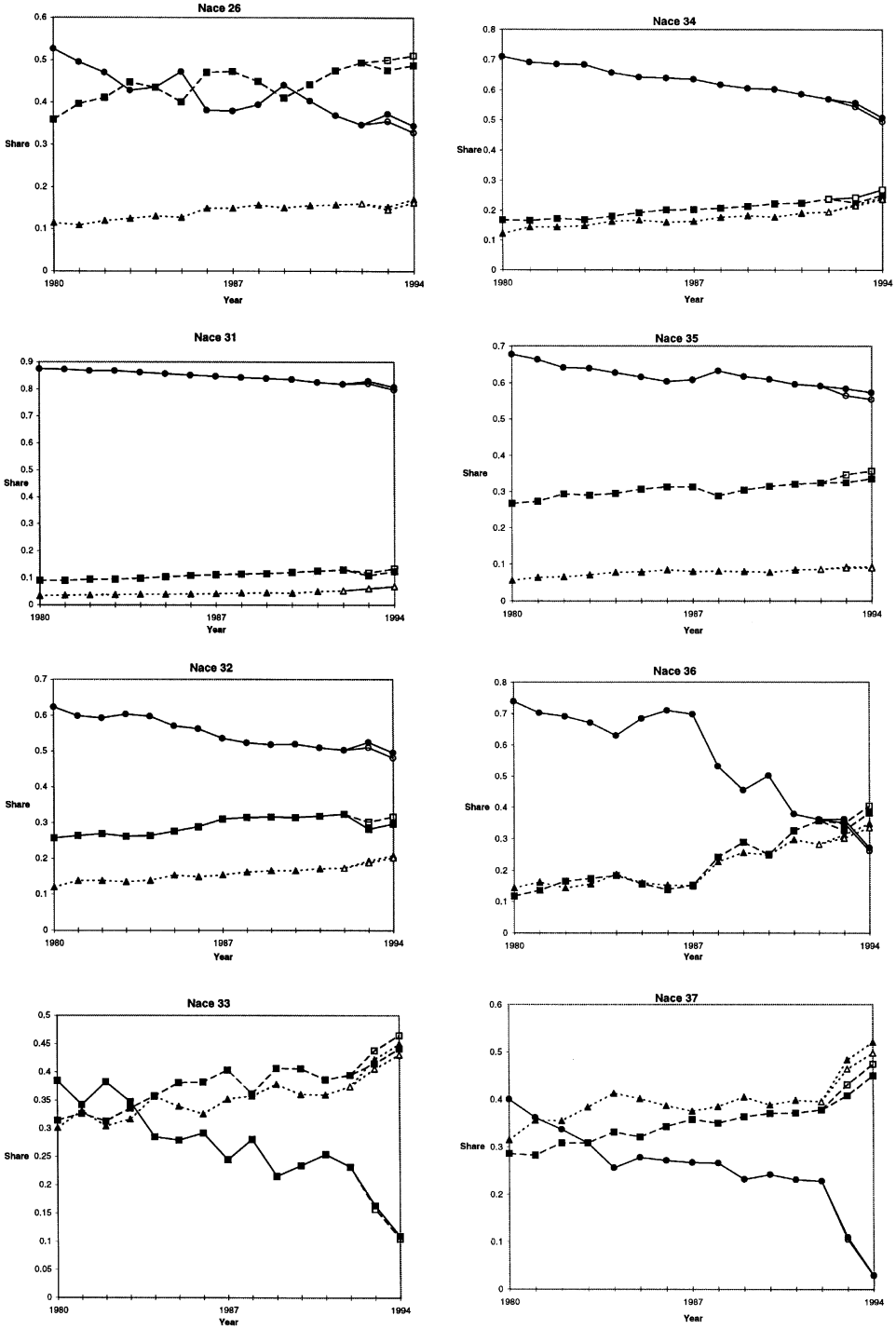


Figure A1. ***Continued***

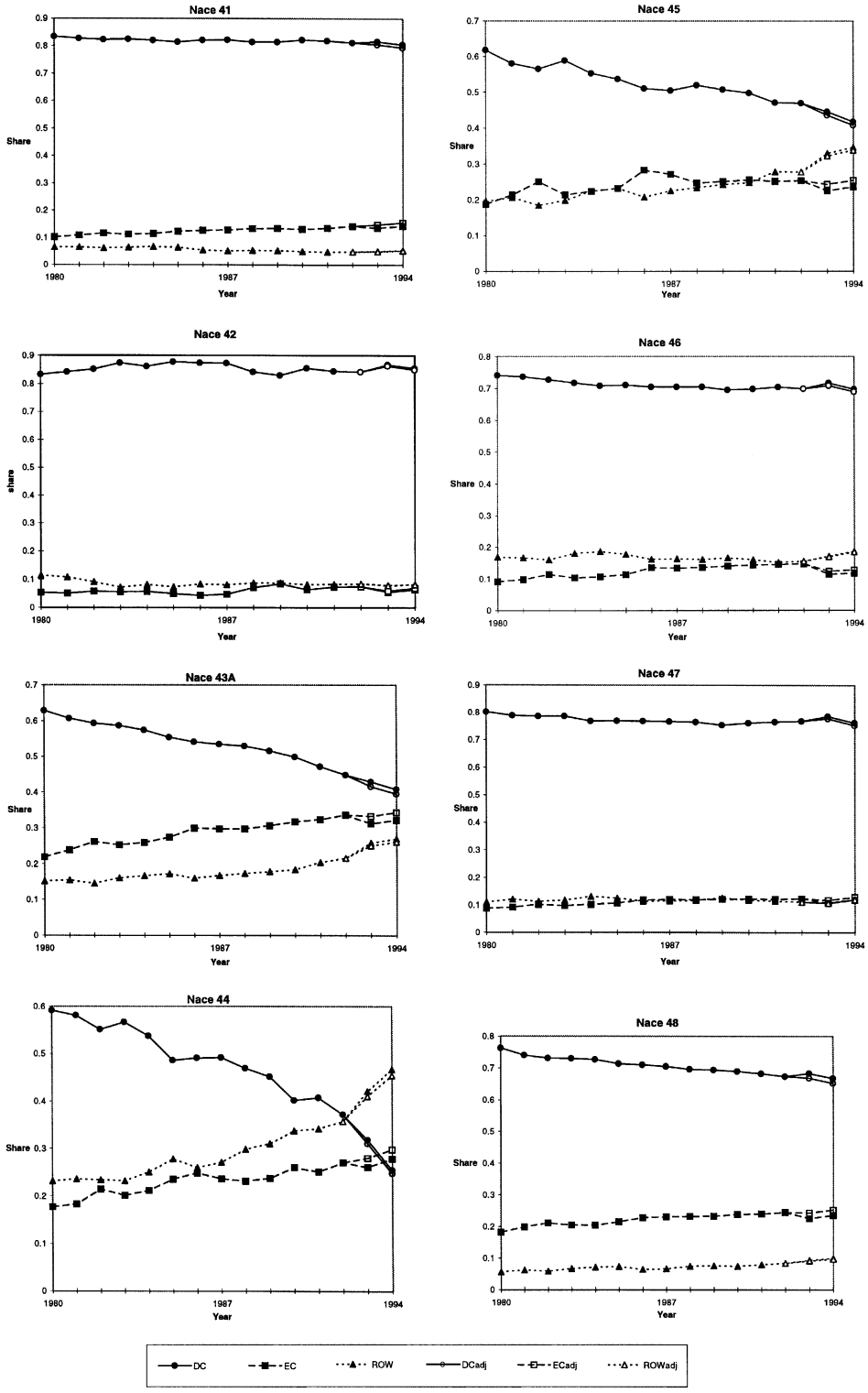


Figure A1. Actual and adjusted changes in traded shares: two-digit Nace

APPENDIX B. THE ECONOMETRIC MODELLING OF THE DEMAND AND SUPPLY EFFECTS

We have followed the methodology of traditional econometric studies of integration in accounting for *demand effects* of the Single Market by adding dummy variables to a well-specified demand system. Earlier work in this tradition includes Winters (1984), Martínez and de Boer (1995) and Brenton and Winters (1992). An Almost Ideal Demand system (Deaton and Muellbauer, 1980) is estimated to represent consumer preferences over goods differentiated by place of production. Consumers allocate their demand for the different countries' goods depending on (unobserved) relative consumption prices.

We can write the sectoral demand equations for the import and domestic market shares in the form

$$s_{ik} = \alpha_i^* + \sum_{j \neq k} \gamma_{ij} \ln(p_{jk}^m / p_k^d) + \beta_i \ln(Y_k / P_k^*) \tag{B1}$$

where s_{ik} is the market share of the ith country supplier in the kth country's market, and p_{jk}^m and p_k^d respectively are the import price of the jth country importer into the kth country market and the home market price of kth country producers themselves. Y_k is total nominal sectoral expenditure by k residents, and P_k is a Stone sectoral-aggregate price index: $\ln P_k^* = \bar{s}_{kk} \ln p_k^d + \sum_{j \neq k} \bar{s}_{jk} \ln p_{jk}^m$, where the bar denotes base-period shares. The adding up, price homogeneity and Slutsky symmetry properties of the demand system are imposed on our estimates. Global negativity can be imposed only at the cost of severe restrictions on the possible range of price elasticities (Diewert and Wales, 1987), and has therefore been imposed only if local negativity would have otherwise been violated.

The demand effects of the Single Market measures are modelled as affecting the mark-up between the actual (or shadow) consumption prices of imported goods and the price of those goods in their own home market. We can write the typical equation for the consumption price of the jth country's imports into the kth market as

$$\ln p_{jk}^m = \ln p_j^d + \eta_{jk} + \tau_{jk} \tag{B2}$$

where p_j^d is the domestic producer price in the jth country and η_{jk} is the pre-Single Market mark-up of the consumption prices of imports into the kth market over their price in their home market. τ_{jk} is a dummy variable that measures the change in this mark-up as a result of the Single Market programme.

Substituting these import price equations into the demand share equations (B1), we obtain an expression for the ith budget share in the kth country in terms of relative domestic and importer firm home market prices:

$$s_{ik} = \alpha_i + \sum_{j \neq k} \gamma_{ij} \ln(p_{jk}^d / p_k^d) + \beta_i \ln(Y_k / P_k) + \tau_i^k \tag{B3}$$

The Single Market dummy terms τ_{jk} will form new composite dummy variables in each of the reduced-form demand equations, $\tau_i^k = \sum_{j \neq k} (\gamma_{ij} - \beta_b \bar{w}_j) \tau_{jk}$. We are unable to identify individually the parameters η_{jk} and τ_{jk}. The η_{jk} terms will be incorporated into the constants of the equations: $\alpha_i = \alpha_i^* + \sum_{j \neq k} (\gamma_{ij} - \beta_b \bar{s}_{jk}) \eta_{jk}$. The sectoral-aggregate price index will now be defined solely in terms of domestic prices: $\ln P_k = \bar{s}_{kk} \ln p_k^d + \sum_{j \neq k} \bar{s}_{jk} \ln p_{jk}^d$. The set of reduced-

form demand equations will continue to inherit the properties of rationality of the original set of equations (B1) under the same set of restrictions as discussed above.

Our initial estimates were of the partial adjustment system with a diagonalized adjustment matrix. The requirements of consumer theory, homogeneity and symmetry were imposed to ensure the efficiency of the estimates. Time trends were initially excluded from the estimates. For the estimation we used maximum likelihood, jointly estimating the set of demand equations. We have estimated a full covariance matrix, since we have relatively low dimensionality and the use of a misspecified model would be worse. Tests were undertaken for autocorrelation, ARCH processes and non-normality of residuals, and further specification searches were pursued if necessary. In our empirical estimates, we allow a dummy variable in our dynamic equations for each year from 1992 onwards. The overall Single Market impact is then determined by evaluating the permanent effect on market shares, taking account of equation dynamics.

We have gone beyond previous empirical work on market integration by also modelling the *supply effects* of the Single Market Programme on price–cost margins. These competitiveness effects were identified by Smith and Venables (1988) and Emerson *et al.* (1988) as playing an important role in the overall gains from the Single Market.

Our approach to modelling supply is to examine the determination of domestic prices within an explicit oligopoly model developed by Allen (1994). Domestic firms are assumed to compete directly with import competitors in setting their prices. Consumer demands are given by an Almost Ideal Demand system. In general, the more competitive product markets are, the more price–cost margins will be forced to narrow. Hence we can obtain evidence on changes in competitiveness on domestic price setting as a result of the Single Market, by including dummy variables in these price equations.

Firms maximize profits, taking into account the conjectural variation of their expectation of their competitors' responses to their strategy. The first-order conditions for profit maximization are therefore

$$\frac{d\Pi_i}{dp_i} = \frac{\partial \Pi_i}{\partial p_i} + \sigma_i \sum_{j \neq i} \frac{\partial \Pi_i}{\partial p_j} \frac{p_j}{p_i} = 0 \tag{B4}$$

where Π_i is the ith firm's profit.

Allen (1994) shows that, with demands given by an Almost Ideal Demand system, a linearized pricing equation can be derived of the following form:

$$\ln p_i = \frac{-\alpha_i}{\gamma_{ii}(2-\sigma_i)} + \frac{(1-\sigma_i)}{(2-\sigma_i)} \ln c_i' + \frac{1}{(2-\sigma_i)} \sum_{j \neq i} \frac{-\gamma_{ij}}{\gamma_{ii}} \ln p_j \tag{B5}$$

where α_i and γ_{ij} are parameters of the demand system, and c_i' are marginal costs.

We specify the technology of firms in country j by a Cobb–Douglas cost function. Marginal costs are then given by

$$c_j' = w_j^{\alpha} m_j^{1-\alpha} y_j^{\delta} \tag{B6}$$

where w_j is labour costs, y_j is the cost of other inputs, and y_j is the output of the industry. The parameter δ is a measure of the returns to scale.

We have examined a very general specification for the price equation. In general, prices

are a mark-up on both domestic marginal costs and competitor import prices:

$$\ln p_{ij} = \mu_j(\alpha \ln w_j + (1-\alpha)\ln m_j + \delta \ln y_j) + (1-\mu_j)\sum_{i \neq j} \zeta_{ij} \ln p_{ij} + \varepsilon_j + \sigma_j \tag{B7}$$

The parameters ζ_{ij} are a set of weights on importers' prices. They can in principle be explicitly derived from the parameters of the demand function. The term ε_j is the mark-up prior to the Single Market Programme and σ_j is a dummy representing the effects of the measures themselves on the mark-up.

Again, to preserve degrees of freedom we have adopted a partial adjustment formulation of the price equations. In order to increase the efficiency of our estimates, price homogeneity has been imposed. For our final estimates, we have used iterative non-linear SURE to estimate the set of different country equations for each sector simultaneously.

To evaluate the *overall impact* of the Single Market legislation on trade flows in the European Union, we have simulated our estimated demand equations to take account of the impact of the supply effects of the Single Market. Such estimates ignore the supply-side effects of the Single Market, which work to reduce domestic prices and costs themselves by increasing competition and enabling the exploitation of economies of scale. In our analysis we have attempted to measure these effects through a further set of dummy variables, σ_j, which are estimated as part of the set of domestic price equations.

APPENDIX C. COMPUTABLE GENERAL EQUILIBRIUM MODELLING METHODOLOGY

Consider first final demand. p_{ij}^k and x_{ij}^k denote the price and quantity of a single product variety of industry k produced in country i and used (as a final demand) in country j. (There are $n_i^k m_i^k$ such varieties, and, because of symmetry, we do not need to introduce a notation for individual varieties.) We follow Dixit and Stiglitz (1977) in assuming consumer preferences with the following two-stage aggregation property. First, varieties within an industry and country of sale aggregate into a quantity index X_j^k with associated price index P_j^k. The aggregator function has constant elasticity of substitution (common to all countries) denoted by ε_k. Aggregation is over products from all sources; as contrasted with the Armington assumption of separate nesting of products by geographical source. Second, the price indices, P_j^k, enter the expenditure function of a single representative consumer with homothetic preferences in each country. If u_j is utility, E_j is the unit expenditure function and M_j is income, then the budget constraint of the country j consumer is

$$M_j = u_j E_j(\ldots, P_j^k, \ldots) \tag{C1}$$

E_j is assumed to be Cobb–Douglas. Consumer demands both for the aggregate quantity indices and for individual varieties are derived by partial differentiation of the expenditure function.

The quantity of a single product variety of industry k produced in i and used as an intermediate good in j is denoted y_{ij}^k with price q_{ij}^k. There is two-stage aggregation of intermediates, analogous to that for final products. First, varieties within an industry and country of sale are aggregated into a quantity index Y_j^k with associated price index Q_j^k (again, not separately nested by geographical source). Second, in each country the quantity indices are aggregated into a composite intermediate commodity whose price index in country j is F_j.

This implies that there is a single composite intermediate commodity, so that the proportions in which each industry uses the products of other industries are assumed to be the same.

The costs of a firm in industry k of country i are given by a cost function c_i^k,

$$c_i^k = m_i^k[h^k(z_i^k)G_i^k(w_i^1, w_i^2, w_i^3)] \tag{C2}$$

$$\text{where } z_i^k = \sum_j (x_{ij}^k + y_{ij}^k) \tag{C3}$$

z_i^k is the total output per variety of a country i firm in industry k. The function h^k describes the returns to scale in industry k. Increasing returns to scale means that $h^k(z_i^k)/z_i^k$ is decreasing in z_i^k, and we employ a functional form for h^k that permits decreasing marginal cost as well as decreasing average cost. Notice that this function is not country specific. Furthermore, there are no economies of scope, since c_i^k is linear in m_i^k, and returns to scale are associated with output per variety z_i^k. The function G_i^k aggregates input prices into cost per unit h. Its arguments are the intermediate price index, F_j, and factor prices, w_i^l. The functions G_i^k differ by country, but only by a scalar, implying Hicks neutral technical differences. Input demands, which in equilibrium equal factor supplies, v_i^l, are partial derivatives of these cost functions, so we have

$$v_i^j = \sum_k n_i^k m_i^k h^k(z_i^k) \frac{\partial G_i^k(F_i, w_i^1, w_i^2, w_i^3)}{\partial w_i^l}, \quad (l = 1, \ldots, 3) \tag{C4}$$

The profits of firms are given by

$$\pi_i^k = m_i^k \sum_j (p_{ij}^k x_{ij}^k + q_{ij}^k y_{ij}^k)(1 - \tau_{ij}^k) - t_{ij}^k(x_{ij}^k + y_{ij}^k) - c_i^k \tag{C5}$$

where τ_{ij}^k and t_{ij}^k are respectively the ad valorem tariff and transaction costs of shipping a unit of industry k output from economy i to economy j.

We assume that firms act as quantity competitors in segmented markets. Each firm in industry k and country i then chooses sales in market j, x_{ij}^k, taking as constant the sales of all its rivals in each market. Optimization requires the equation of marginal revenue to marginal cost in each market, where the slope of each firm's perceived demand curve depends on the extent of product differentiation, and on the share of the firm in that market. Firms' choice of intermediate sales quantities, y_{ij}^k, is less straightforward. It is possible that purchasers of inputs have some monopsony power, to be combined with the monopoly power of sellers. Further, and perhaps more importantly, even if purchasers of intermediates are input price-takers, the demand for intermediates is a derived demand, and establishing the elasticity of the derived demand curve is not straightforward. For these reasons we assume that the price of a good sold as an intermediate equals the price of the same good sold to final demand. Furthermore, the number of varieties of intermediate goods entering the price indices Q_j^k is held constant, so abstracting from any variety effects on the users of intermediate goods.

All that remains to complete the description of the model is the determination of income. Income accruing to factor l in economy i is $w_i^l v_i^l$. National income is factor income accruing to the three factors, plus the profits of firms and CET revenue.

APPENDIX D. DATA UNDERLYING THE EXPERIMENT

In setting the experiment, we have direct estimates for fifteen sectors arising from the

Table D1. Information underlying the choice of experiment

Nace	Description	NTB (1)	Vuln (2)	BIL20 (3)	EU20 (4)	ROW20 (5)	Experiment (6)
22	Metals	1.00	0.00				a
23	Mineral extraction	1.00	0.00			10	a
241:244	Clay, cement, asbestos	2.00	0.00			2	a
245:248	Stone, glass, ceramics	1.75	0.50	a			d
251	Basic chemicals	1.00	1.00	a	15		d
255	Paint and ink	1.00	0.00				a
256	Ind. and agric. chemicals	2.00	1.00	a	17		s
257	Pharmaceuticals	3.00	3.00	a	5	14	d
258	Soap and detergents	2.00	0.00		3	12	s
259	Domestic chemicals	2.00	0.00		19		a
260	Man-made fibres	1.00	0.00				a
311:313	Metal manufacture	1.00	0.00				a
314:315	Metal structures, boilers	2.50	1.50	a		9	d
316	Tools and cans	1.00	0.00			11	a
321	Tractors and agric. machinery	2.00	1.00				a
322	Machine tools	2.00	1.00	a			d
323	Textile machinery	2.00	1.00				a
324	Food and chemical machinery	2.00	1.00	a			d
325	Mining and constr. machinery	2.00	1.00	a			d
326	Transmission equipment	2.00	1.00				a
327	Paper and wood machinery	2.00	1.00			13	a
328	Other machinery	2.00	0.00				a
330	Computers and office machinery	3.00	4.00	a	10		d
341	Insulated wires and cables	3.00	2.00	a	7	4	s
342	Electrical machinery	3.00	2.00	a	11		s
343	Electrical equipment	1.00	0.00		11		s
344	Telecoms and measuring equip.	3.00	4.00	a			d
345	Radio and TV	2.00	1.00	a	4		d
346	Domestic electrical appliances	2.00	1.00				s
347	Electric lighting	2.00	1.00			6	a
351	Motor vehicles	2.00	1.00	a			d
352:353	Motor vehicle parts	2.00	0.00			7	a
361:363,365	Ships, rail stock, cycles	2.33	1.67		1	21	s
364	Aerospace	2.00	1.00	a	8		d
37	Instruments	1.50	1.00		6		s
411,420:423	Grains, pasta, bread	1.40	0.40				a
412	Other foods	1.00	0.00				a
413	Meat products	1.00	0.00				a
414	Dairy products	1.00	0.00				a
415	Fruit and vegetable products	1.00	0.00				a
416:419	Fish products	1.50	0.50		9	20	s
424:428	Drinks	2.50	2.25	a	18	3	d
429	Tobacco	1.00	0.00				a
43A,B	Wool, cotton, silk, flax	1.50	0.50	a		17	a
436	Knitting	2.00	0.00			15	a
438	Carpets	2.00	1.00				a
439	Misc. textiles	2.00	0.00				a
441	Leather tanning	1.00	0.00		13	8	a
442	Leather products	1.00	0.00		20	18	a
451	Footwear	2.00	1.00				a
453	Clothing	2.00	1.00	a			d
455:456	Household textiles, fur	1.50	0.50		2	1	s

continued

Table D1. ***Continued***

Nace	Description	NTB (1)	Vuln (2)	BIL20 (3)	EU20 (4)	ROW20 (5)	Experiment (6)
461:462	Wood boards	1.50	0.00				a
463:465	Other wood	1.33	0.00			5	a
466	Cork and brushes	1.00	0.00				a
467	Wooden furniture	1.00	0.00			19	a
471	Paper and pulp	1.00	0.00				a
472	Processed paper	1.00	0.00				a
473	Print and publishing	1.00	0.00				a
481:482	Rubber	2.00	1.00	a	14	16	s
483	Plastics	2.00	0.00				a
491	Jewellery	2.00	1.00				a
492:493	Musical instruments, photos	1.50	0.50				a
494	Toys and sports	2.00	0.00		16		a

econometric analysis. In column (6) of Table D1, these sectors are marked 'd'. For the remaining sectors, we use either the aggregated manufacturing figure derived from the econometric analysis (indicated by 'a'), or the average figure for the sensitive sectors (indicated by 's'). The choice of experiment is based on the following information. Columns (1) and (2) use the same numbers (aggregation aside) as Davies and Lyons (1996) and are derived from the *ex ante* study of Buigues *et al.* (1990). The columns show respectively the extent to which the sector was affected by non-tariff barriers to intra-EU trade (NTB) and the vulnerability of the sector to other SMP measures, particularly with respect to public procurement (Vuln). The former variable ranges between 1 and 3, the latter between 0 and 4. Column (3) includes the twenty most sensitive sectors as identified by Buiges *et al.* with the highest shares of value added (because of the aggregation procedure that we have employed, this covers eighteen of the listed sectors). Columns (4) and (5) of the table indicate the twenty industries that over the period 1991–4 have experienced the largest change in the share of imports from the EU and from the rest of the world, respectively. For these two columns, the industries are ranked according to the size of the change in share. Finally, we have also used information from the detailed studies on changes in price–cost margins and changes in concentration, as reported in the European Commission's *Single Market Review* discussed in section 2.

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